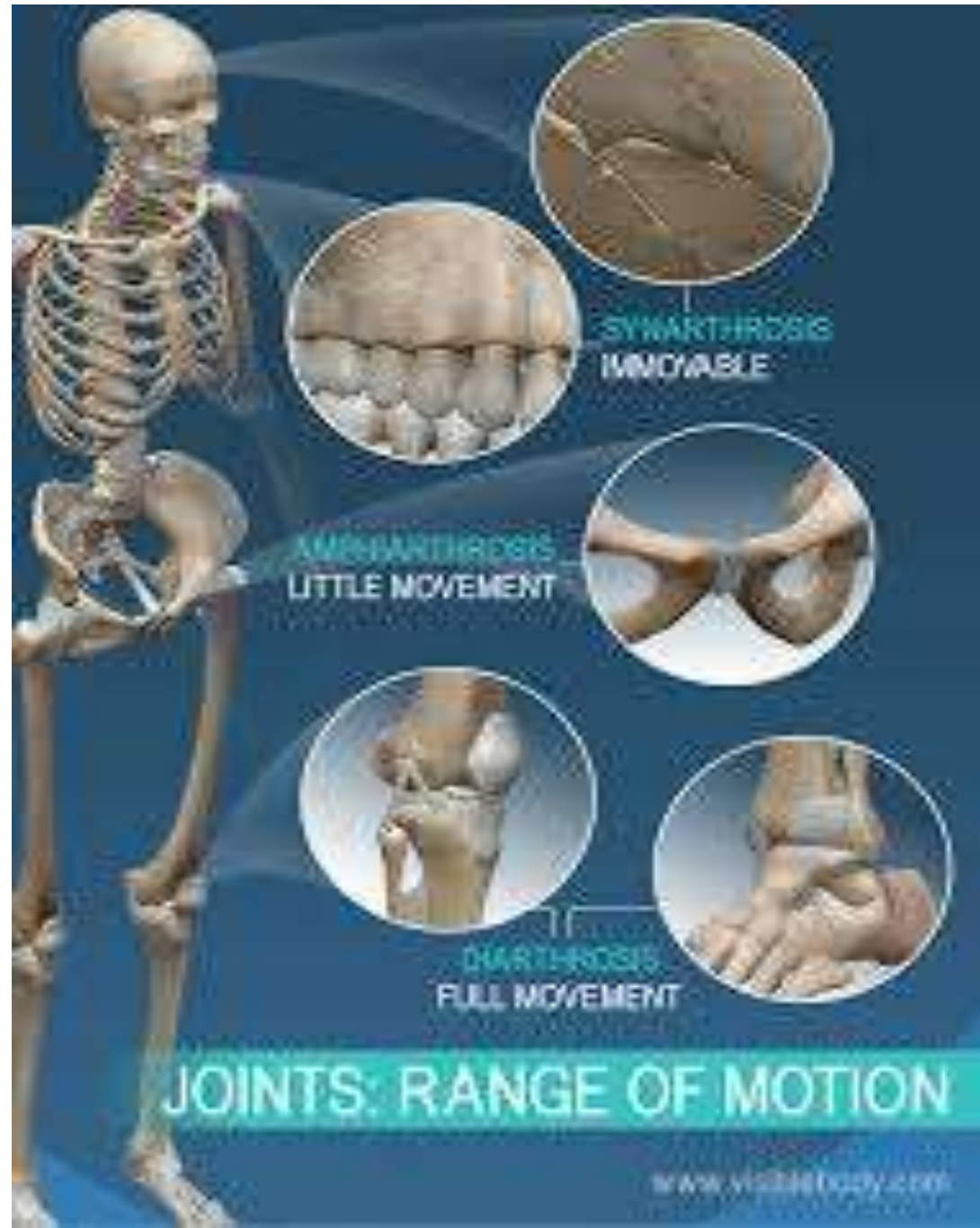


Introduction

Dr. MARIAM WAHBY

Joints: Articulation between two or more bones



Classification of joints:



Fibrous
(Immoveable)



Cartilagenous
(Semi moveable)



Synovial
(freely moveable)

Fibrous joints

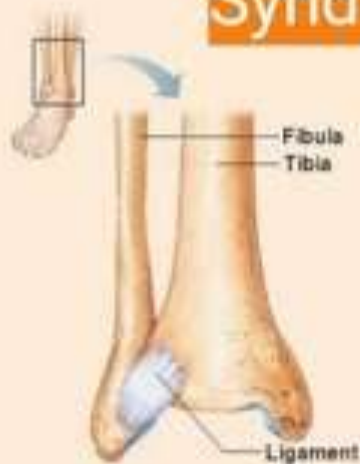
TYPES OF FIBROUS JOINT

Joint Series
2

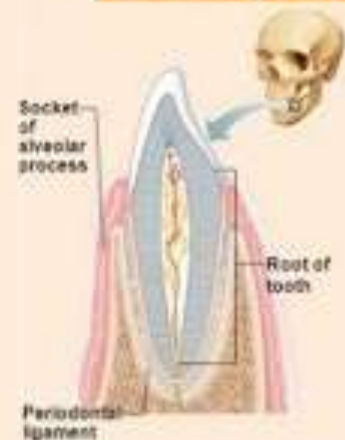
Suture



Syndesmosis



Gomphosis



 **Explanation & Concise notes**

Fibrous joints

Sutures: in skull

Gomphosis: between teeth and gums

syndesmosis: inferior tibiofibular joint.

Cartilaginous joints:

primary cartilaginous joint:

hyaline cartilage

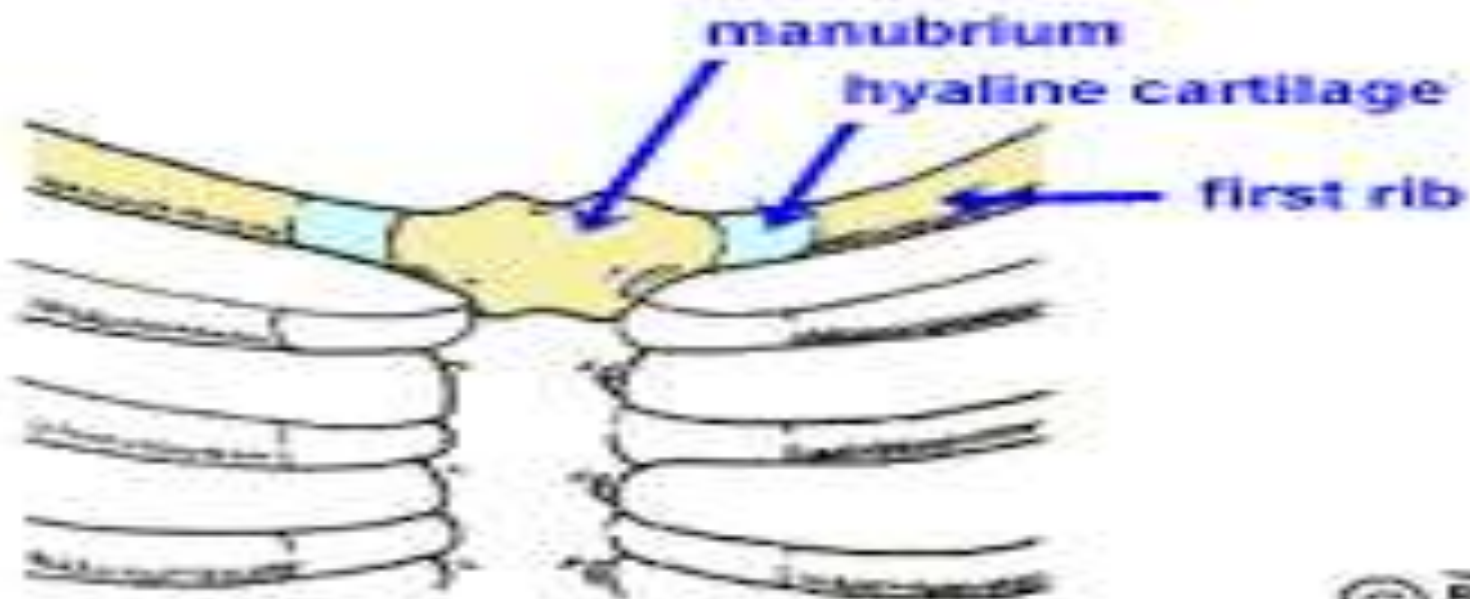
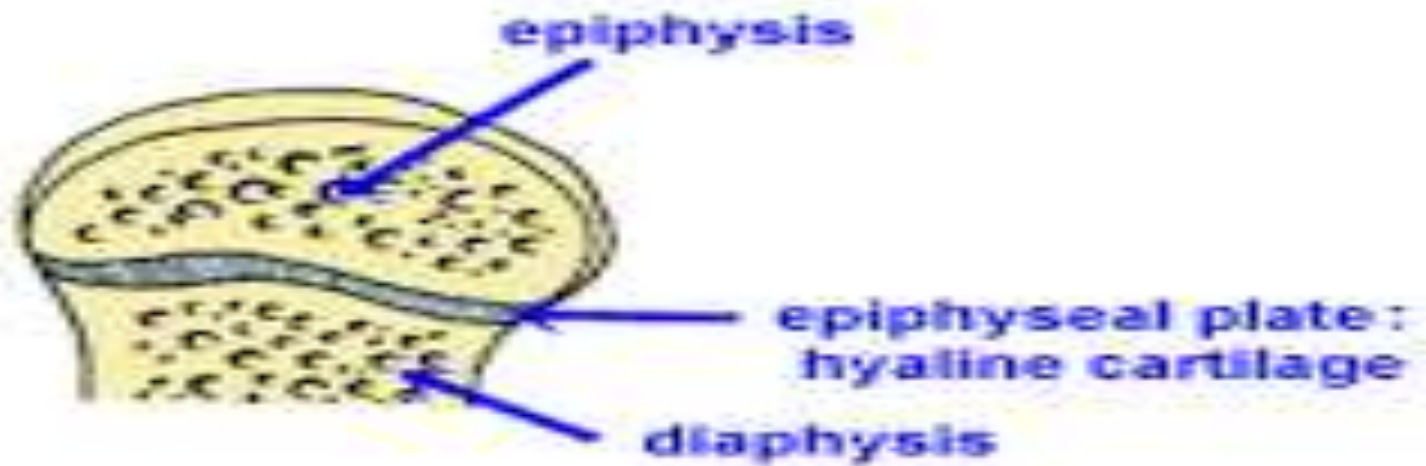
e.g epiphyseal cartilage plate

Secondary cartilaginous joint:

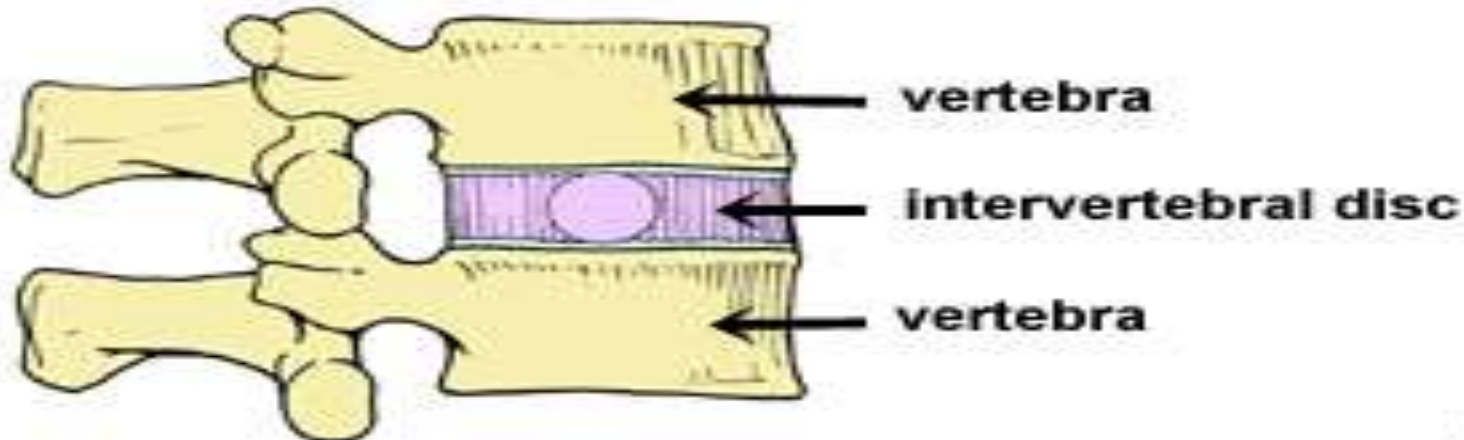
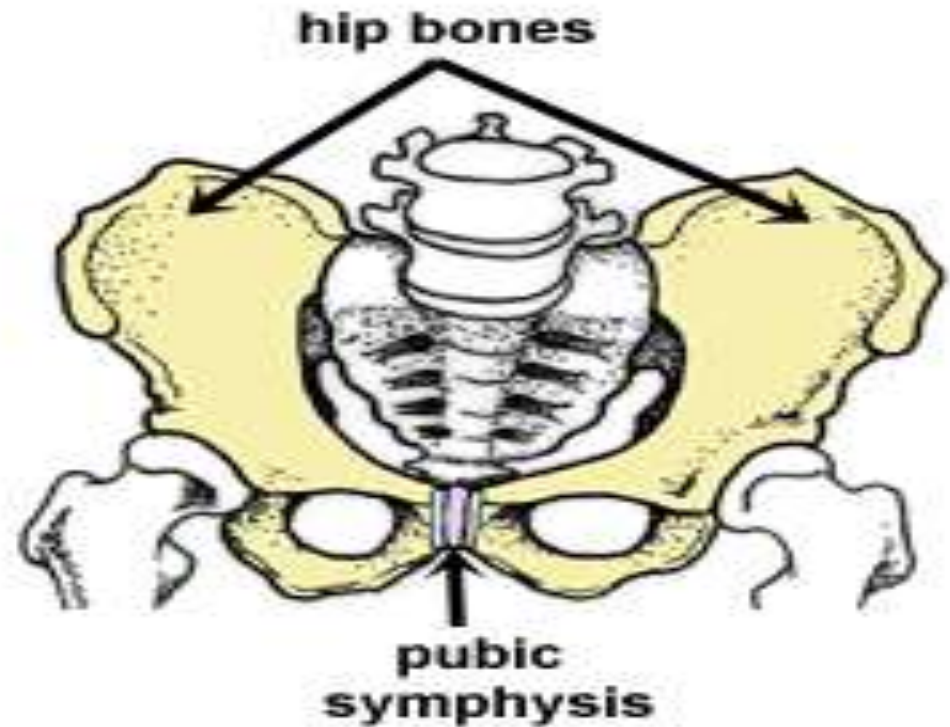
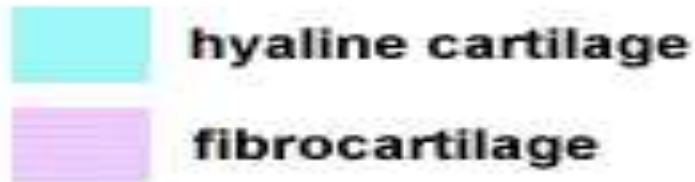
fibrocartilage

e.g intervertebral disc symphysis pubis

Primary cartilaginous joints:

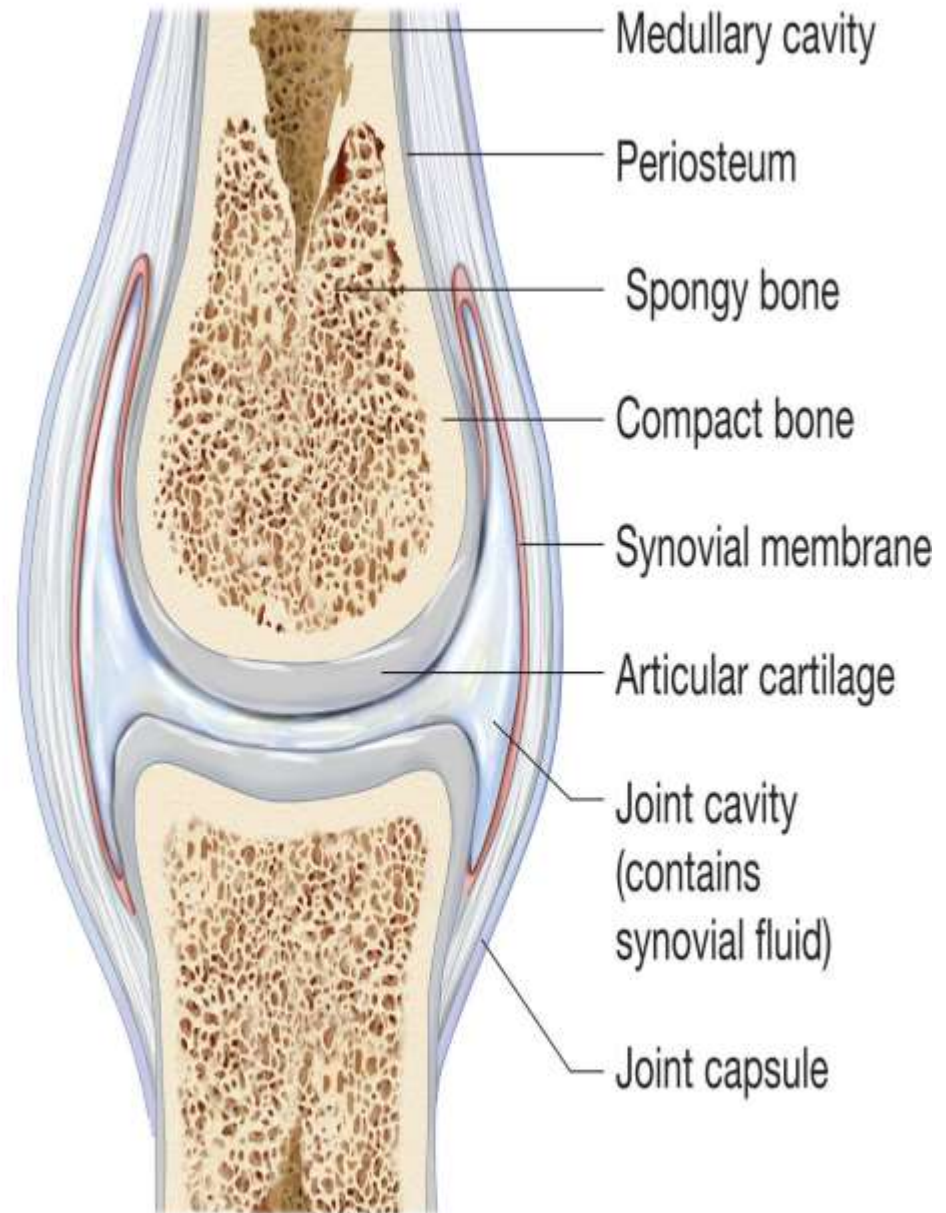


Secondary cartilaginous joints:

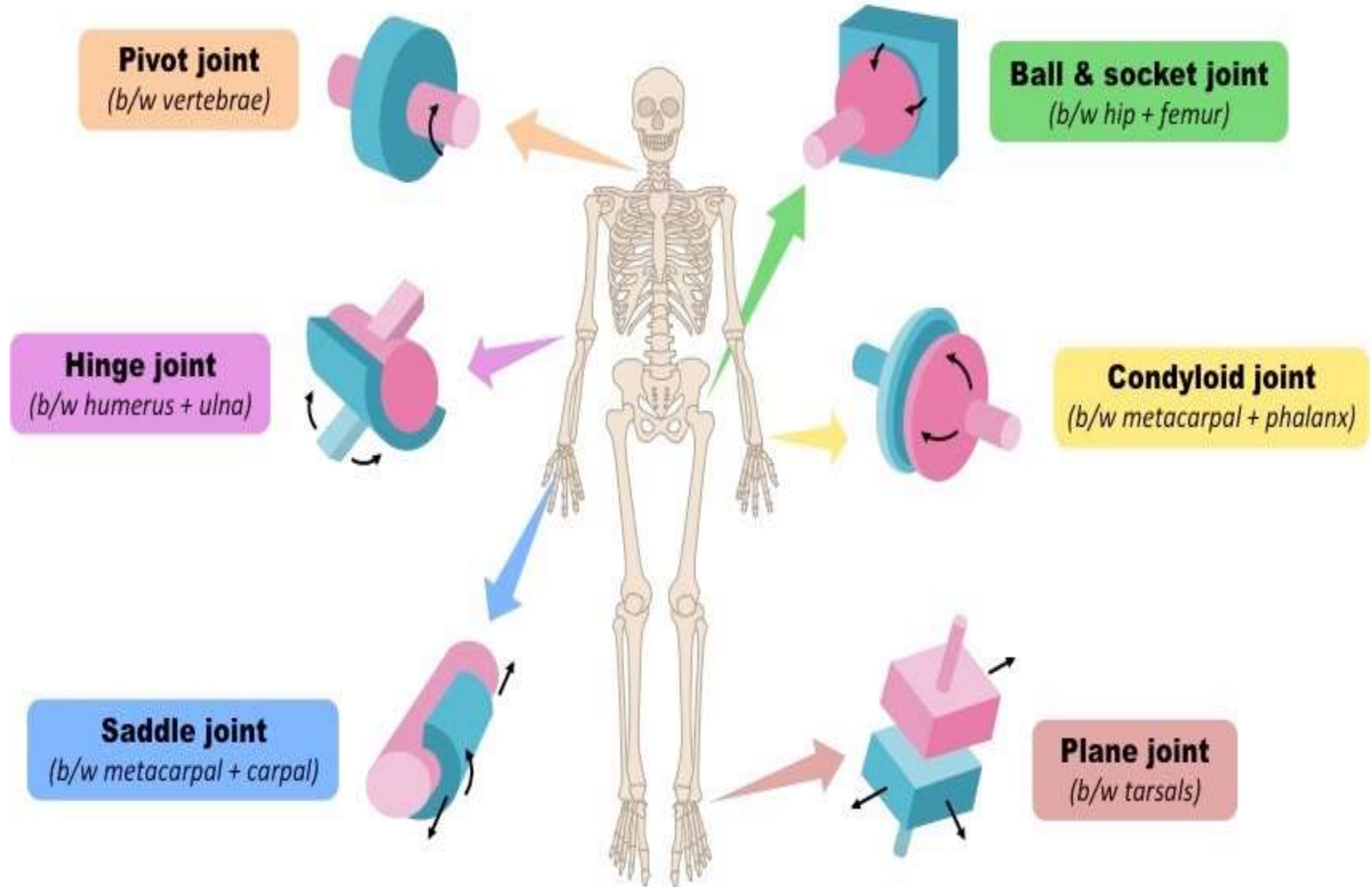


Synovial joint: components

- Bony articular ends
- Articular cartilage
- Fibrous capsule
- Synovial membrane
- Synovial fluid
- Blood and nerve supply



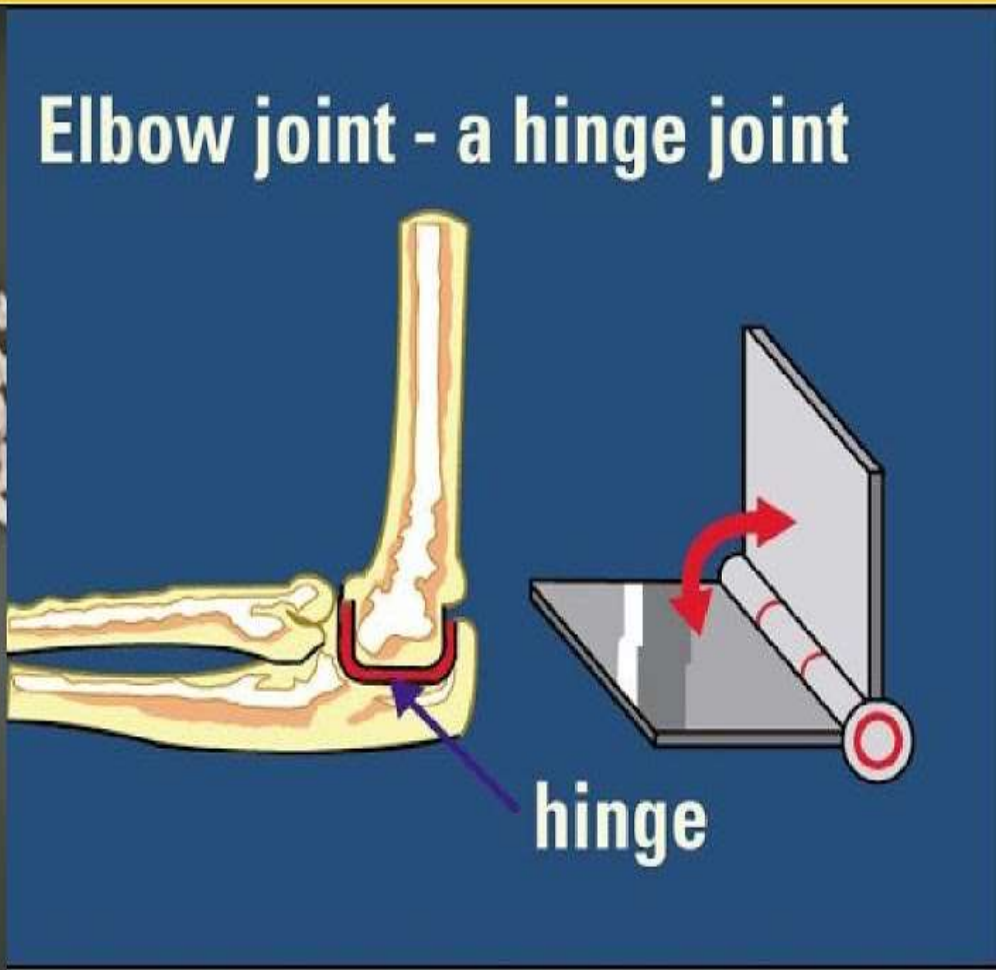
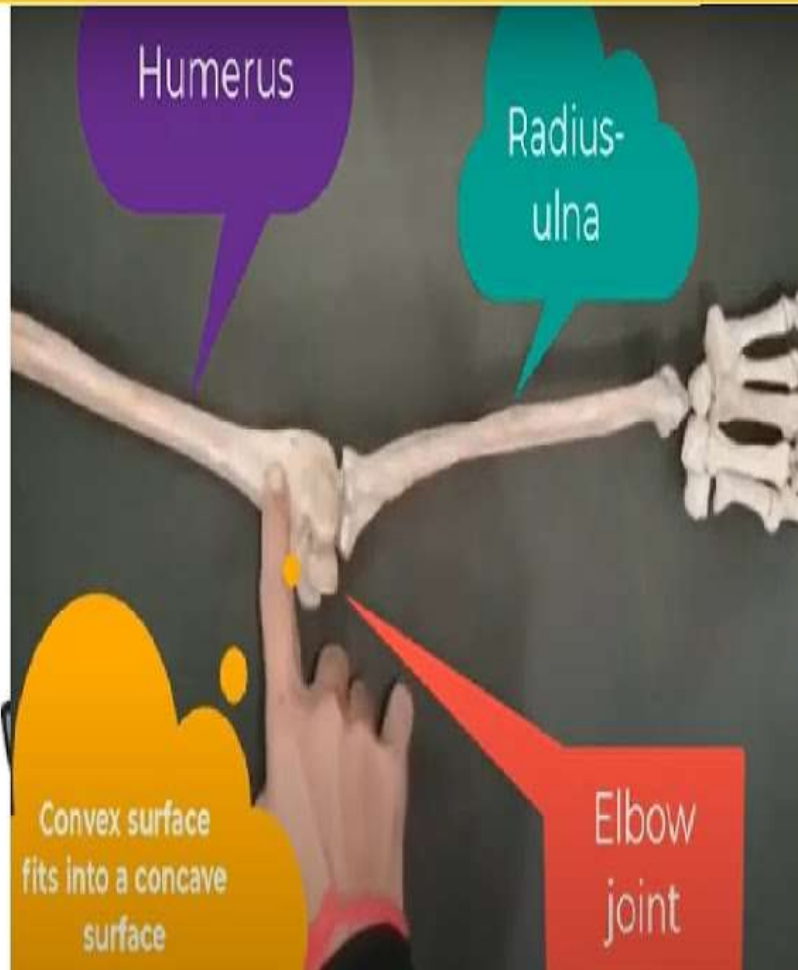
Types of synovial joints:



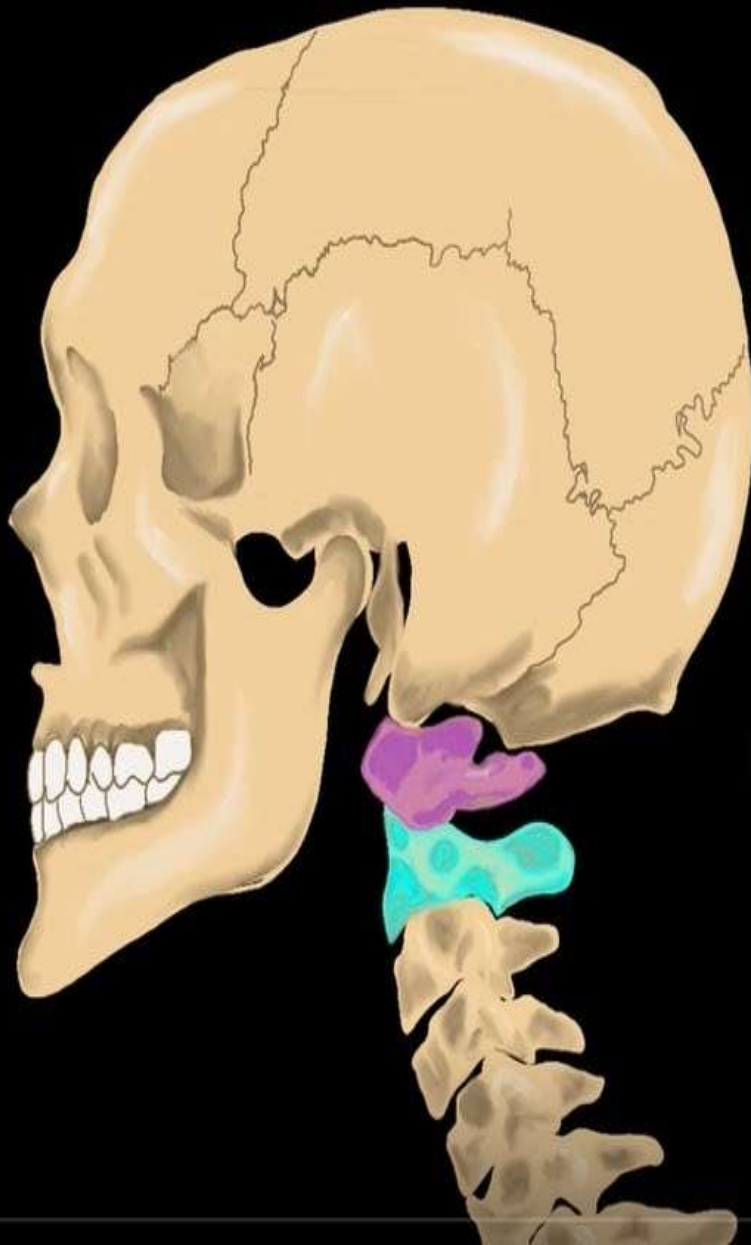


Plane synovial joint
(intercarpal joints)

HINGE JOINT



Pivot Joint

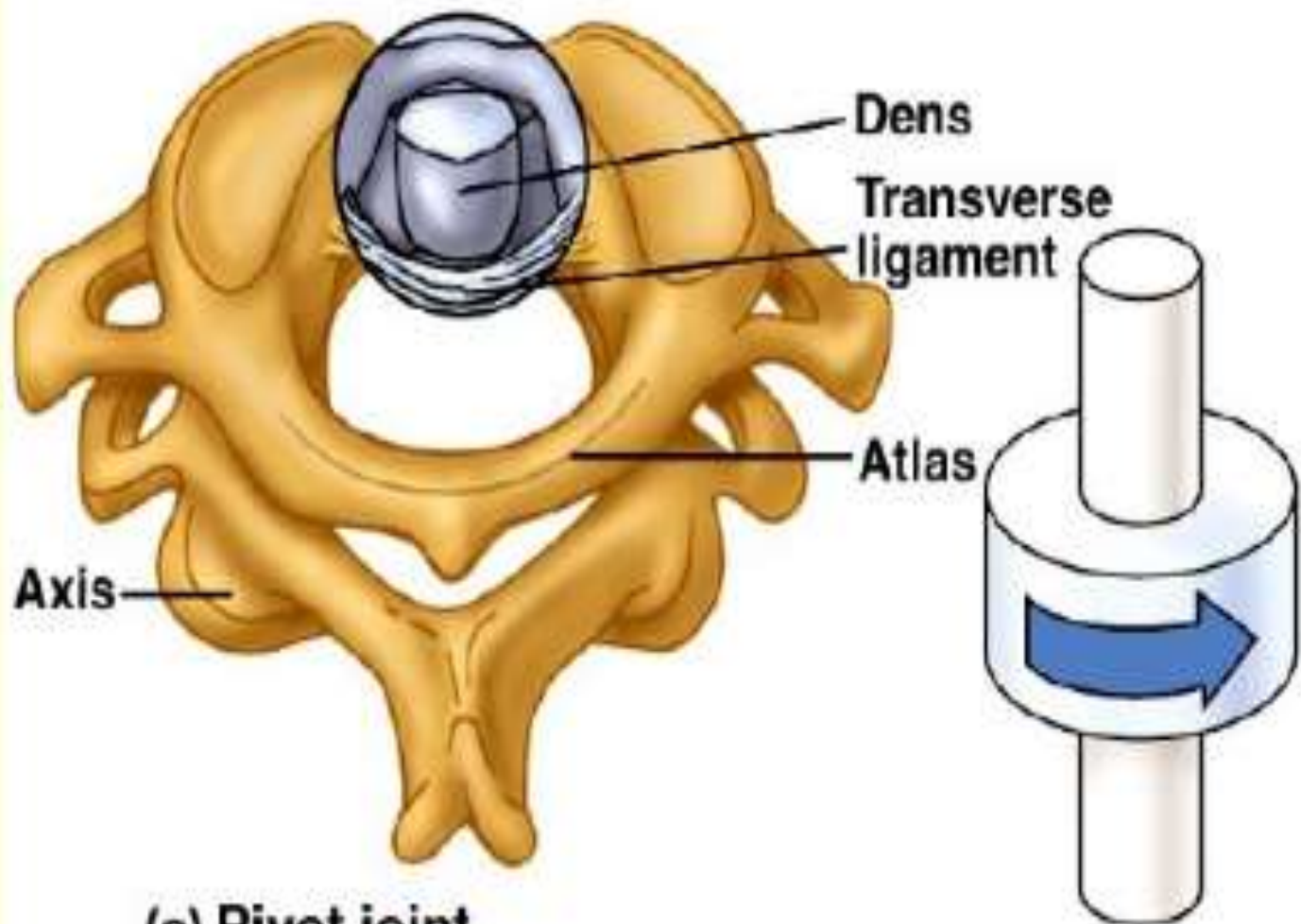


Examples of pivot joint:

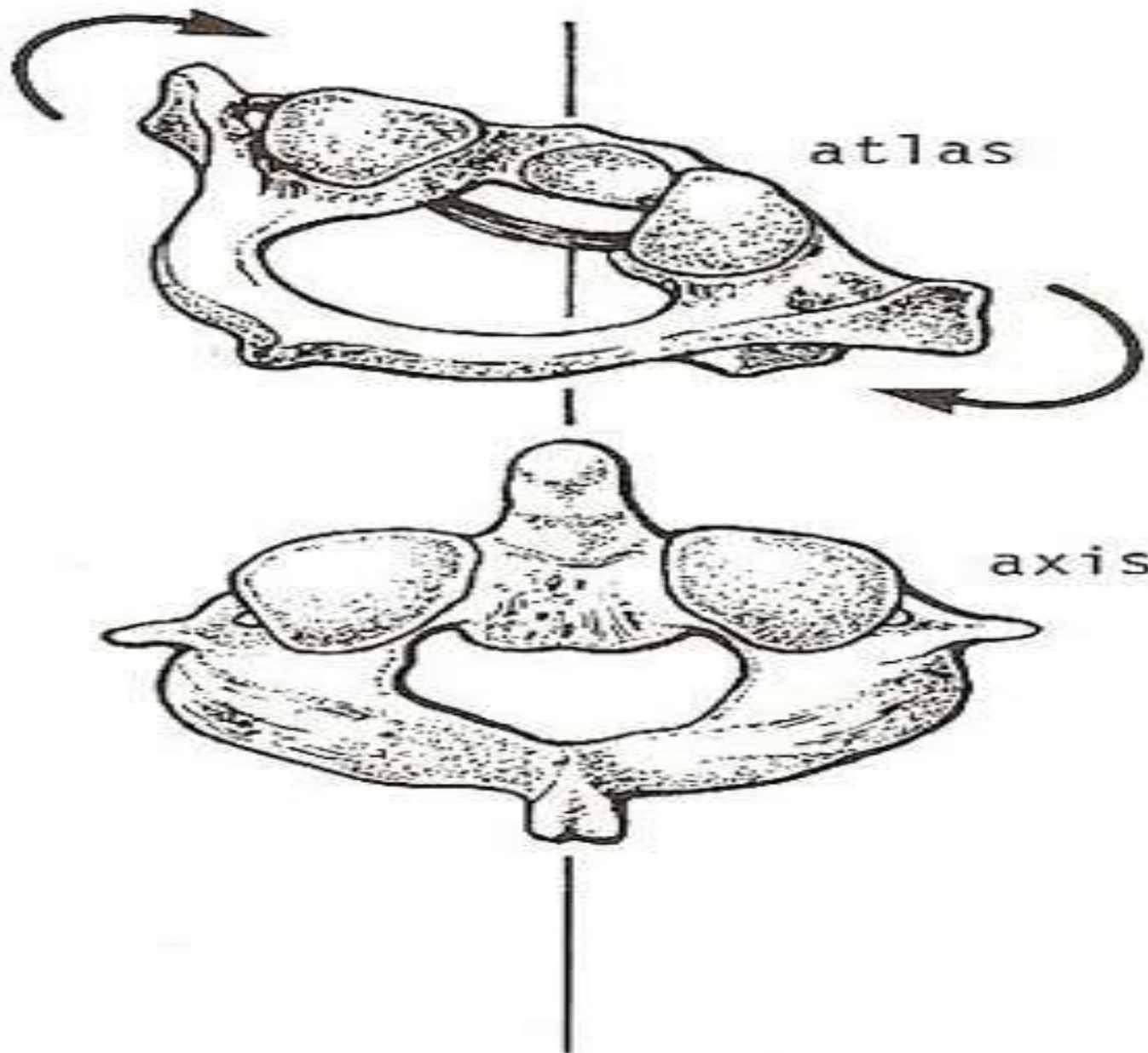
Median atlanto axial joint

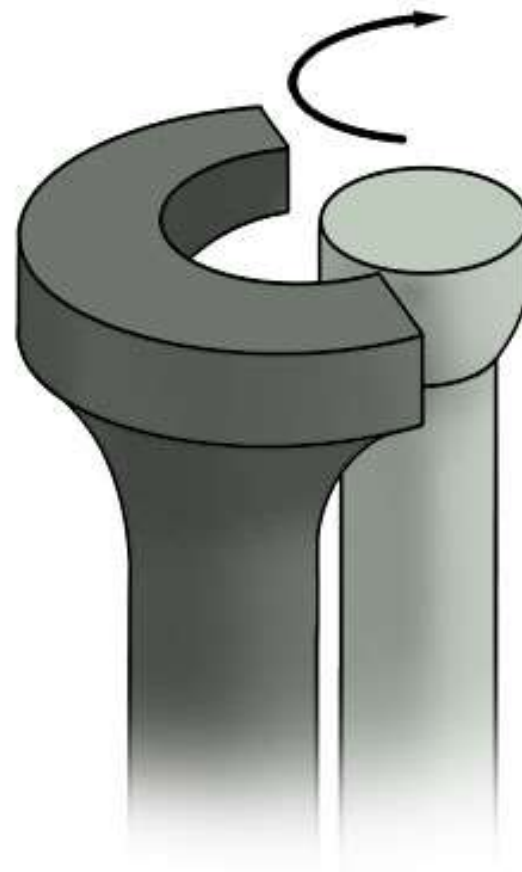
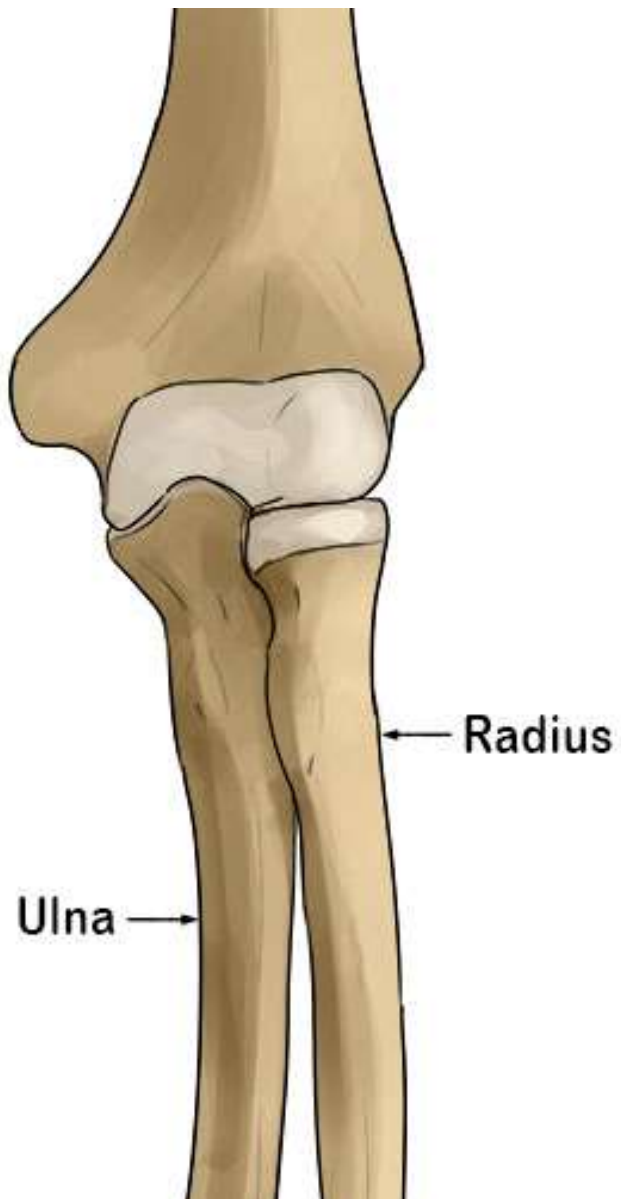
Superior radioulnar joint

- **Pivot joint:**
rotation
around a
central axis
- **Ex:** atlas/
axis joint



(e) Pivot joint





@muscleandmotion

**Wrist
joint**



**Condylloid /
ellipsoid joint**

Condylod joint:

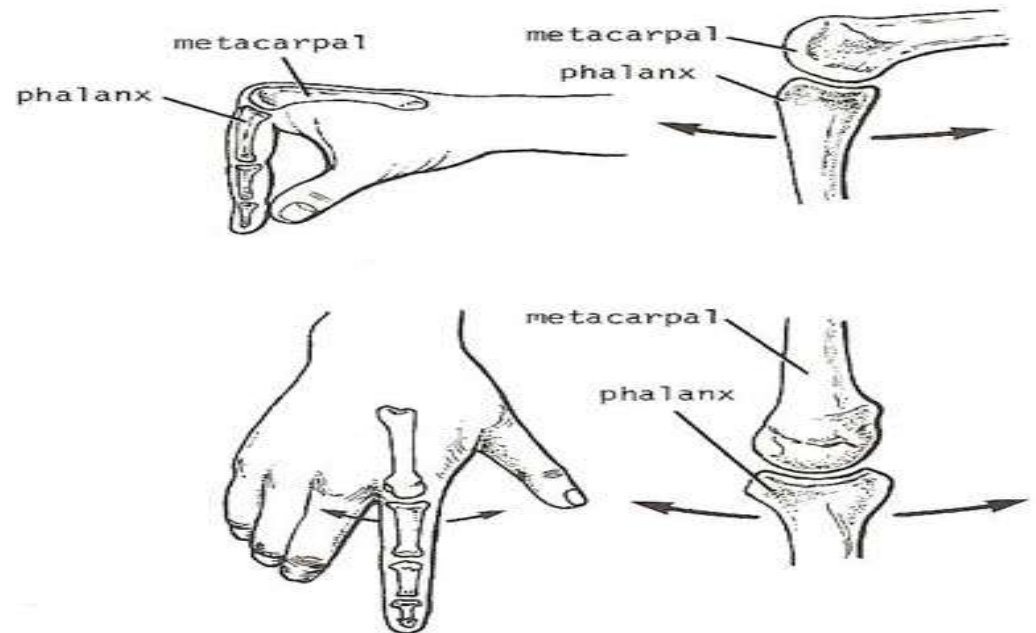
concave surface with convex surface

e.g. metacarpophalangeal (knuckle) joints

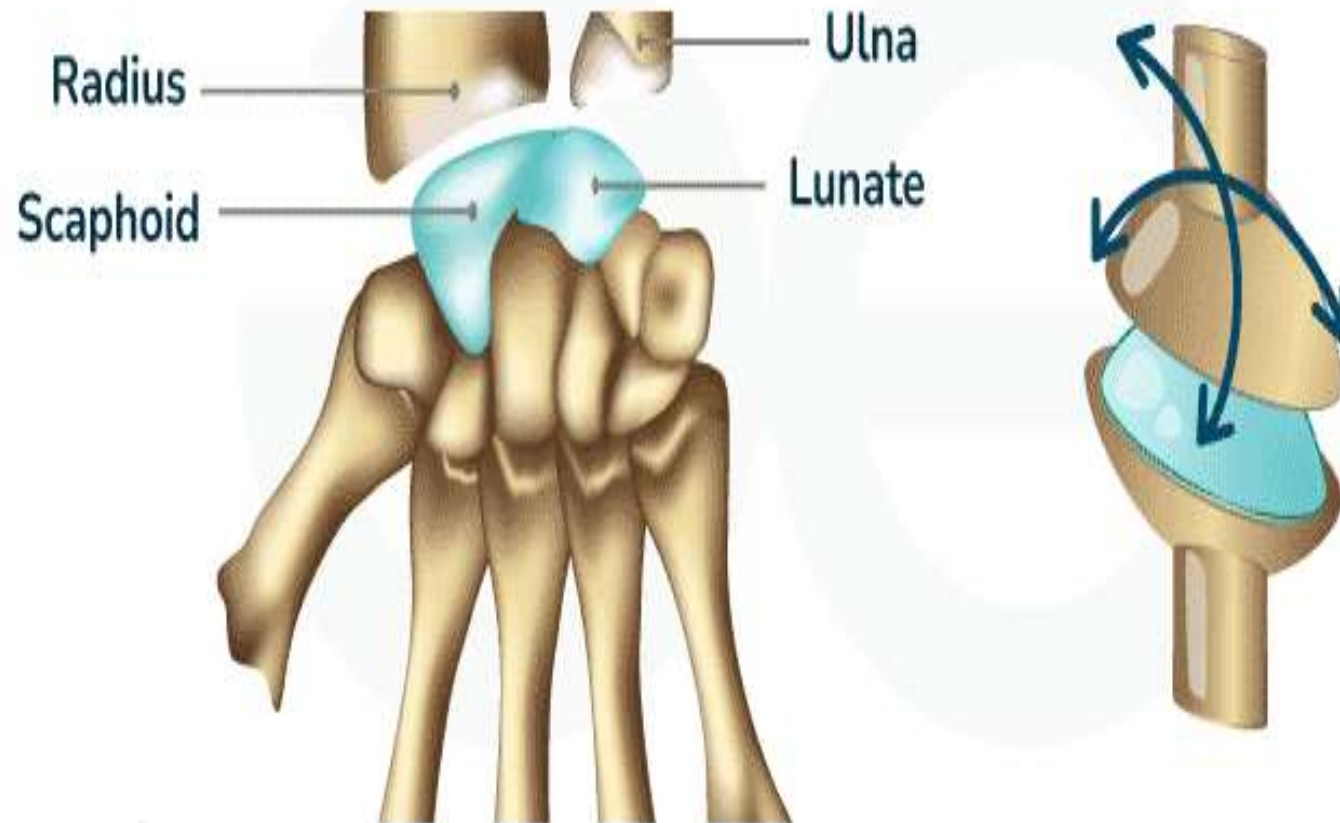
Ellipsiod joint:

Elliptical convex with elliptical concave surfaces

e.g. Wrist joint



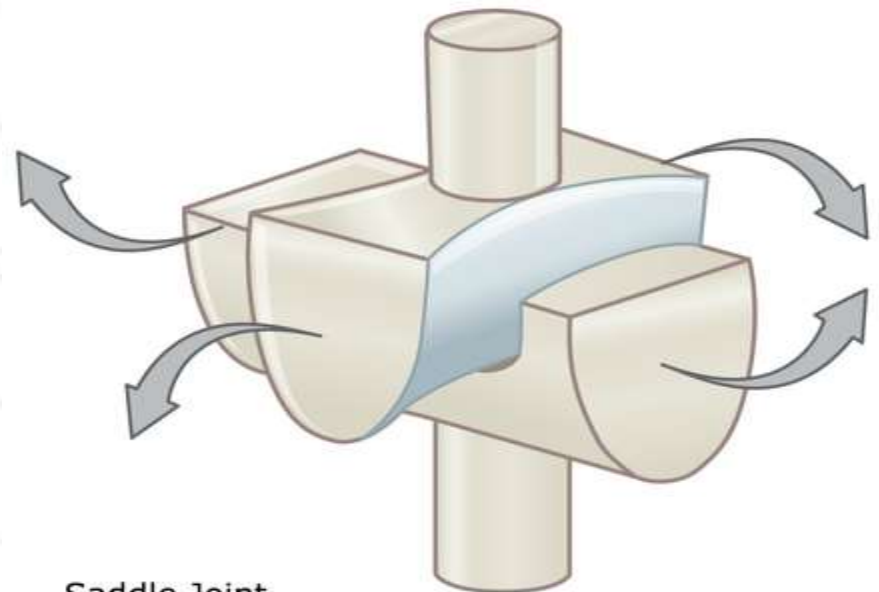
Ellipsoid Joints



Saddle joint:

articular surfaces are reciprocally concavoconvex

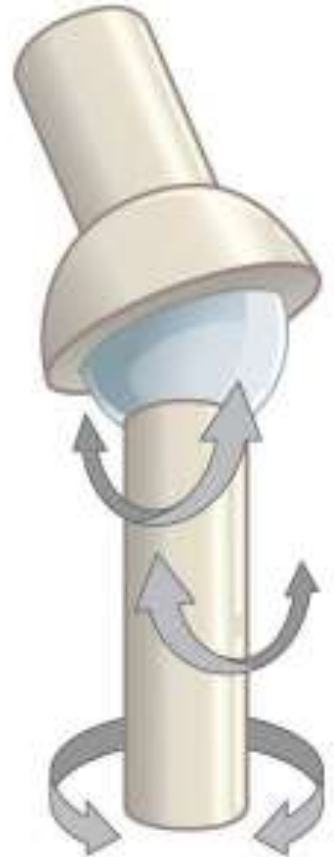
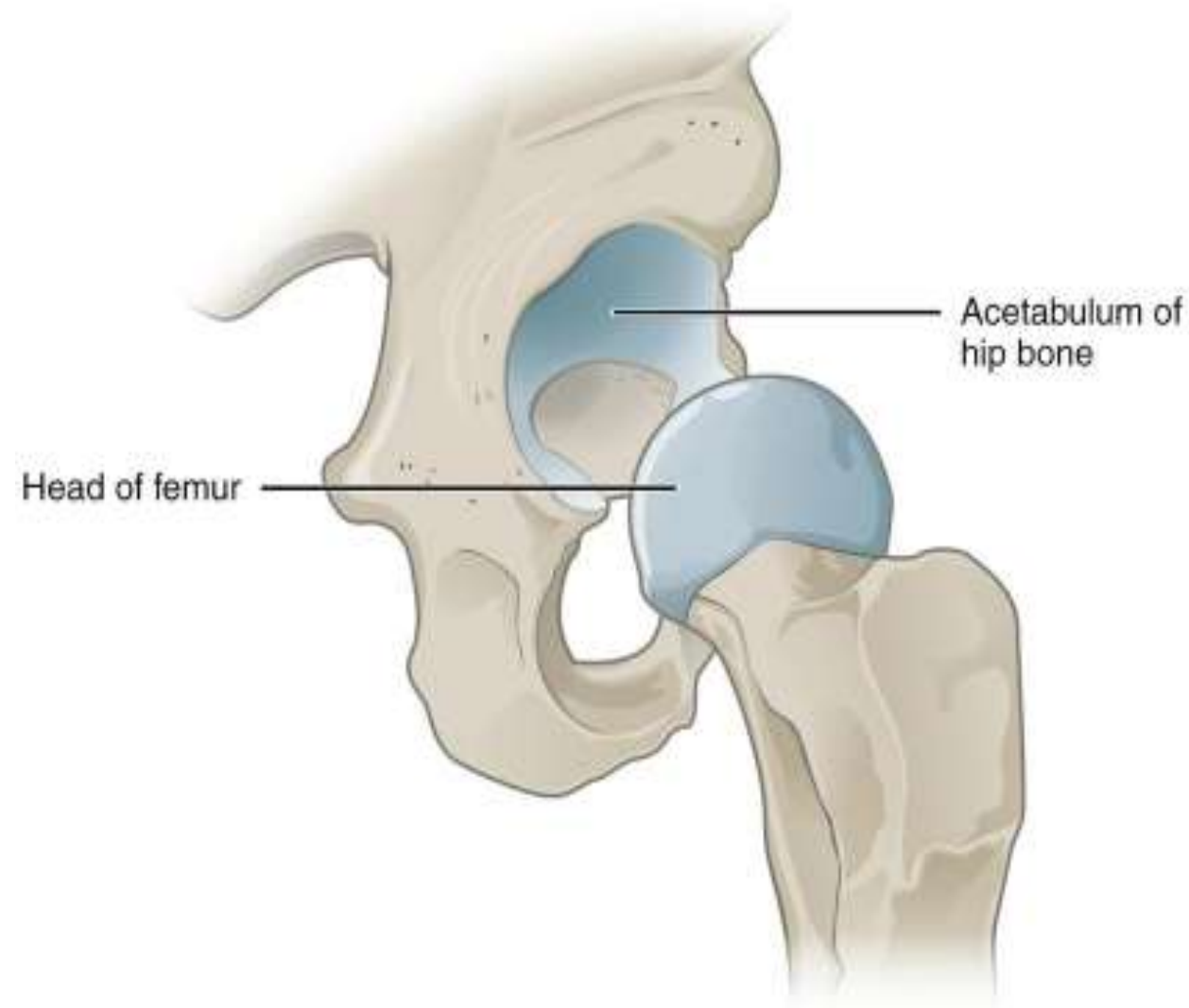
e.g carpometacarpal joint of the thumb



Saddle Joint
eg. CMC Joint of Thumb

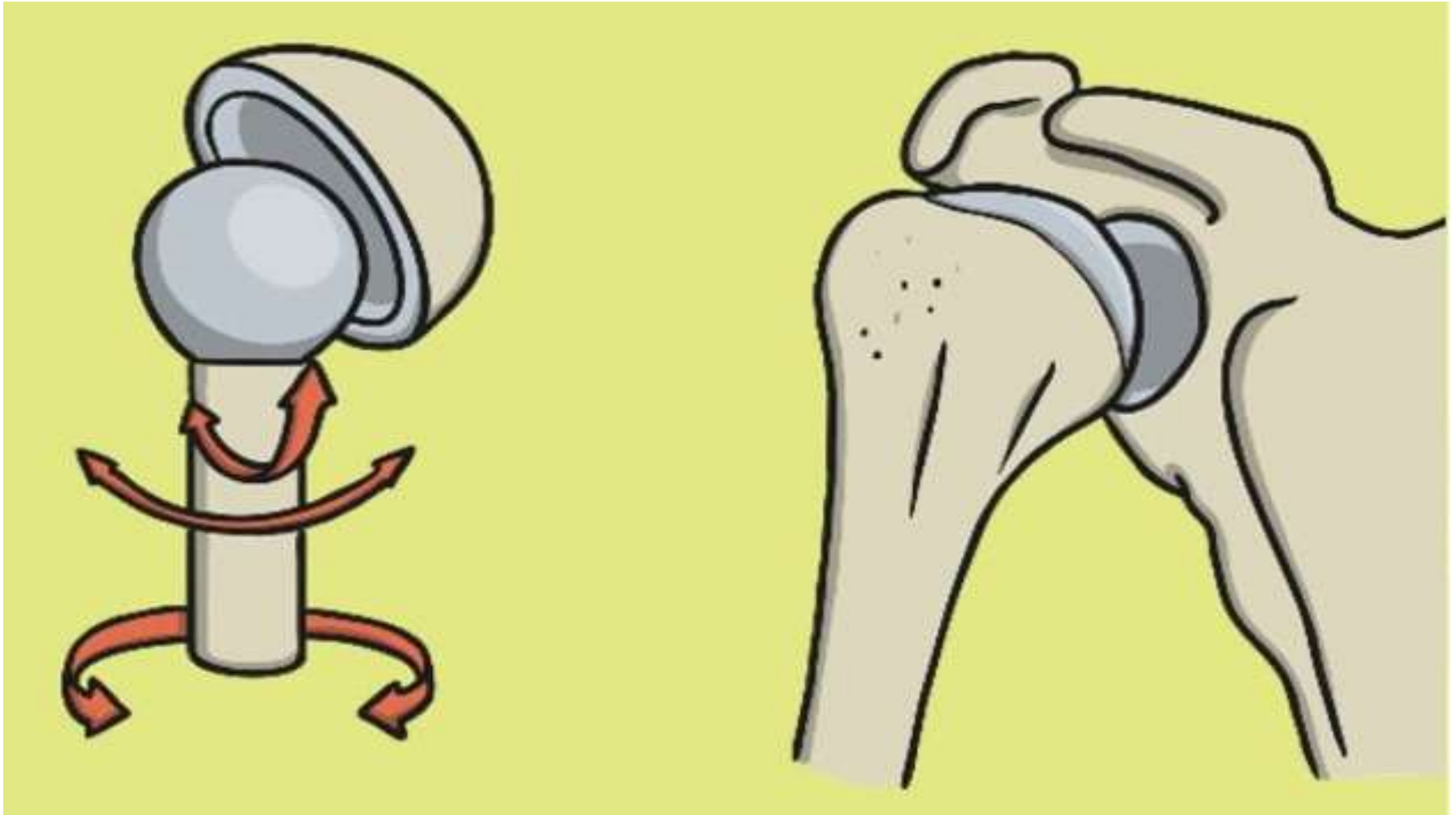
Ball and socket joints:

Hip joint



Ball and socket joints:

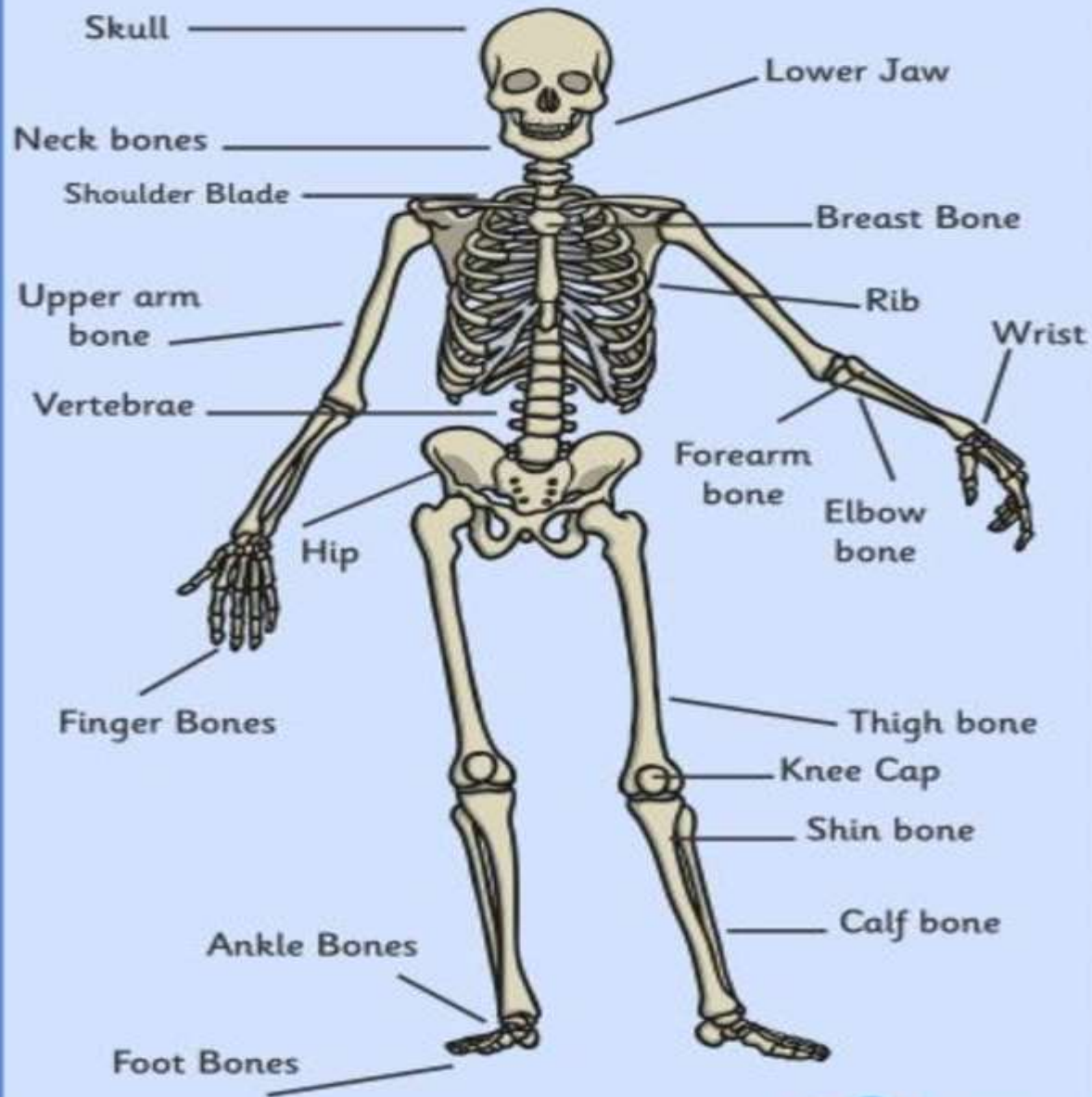
Shoulder joint



Factors Influencing Synovial Joint Stability

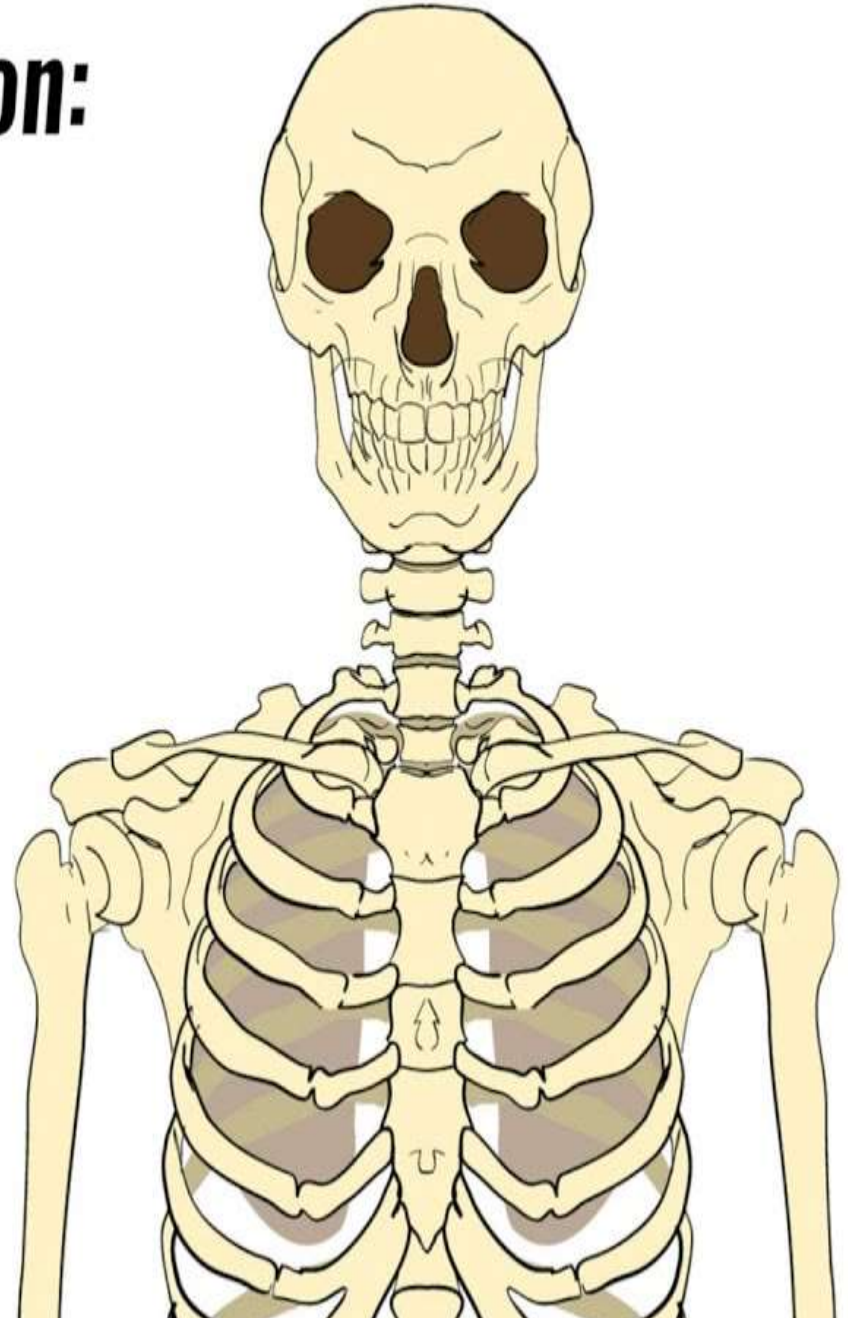
- **The stability of a synovial joint depends on three factors . . .**
 - **The nature of the articular surfaces**
 - **The number and positioning of the ligaments**
 - **The tone and strength of the muscles acting upon the joint**

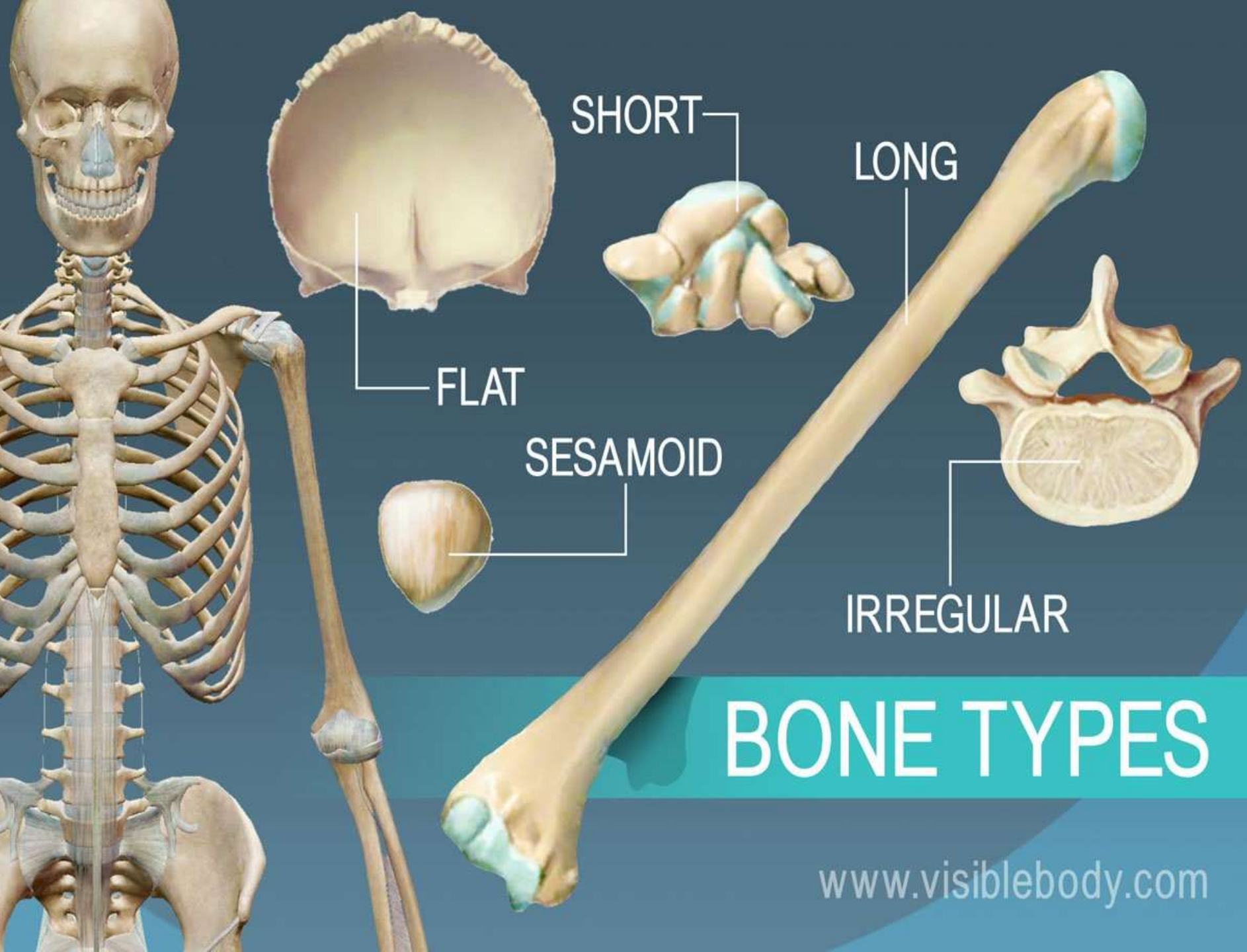
The Human Skeleton



Functions of the skeleton:

- Shape & structure***
- Protect organs***
- Muscle attachment***
- Produce blood cells***
- Calcium storage***





SHORT

LONG

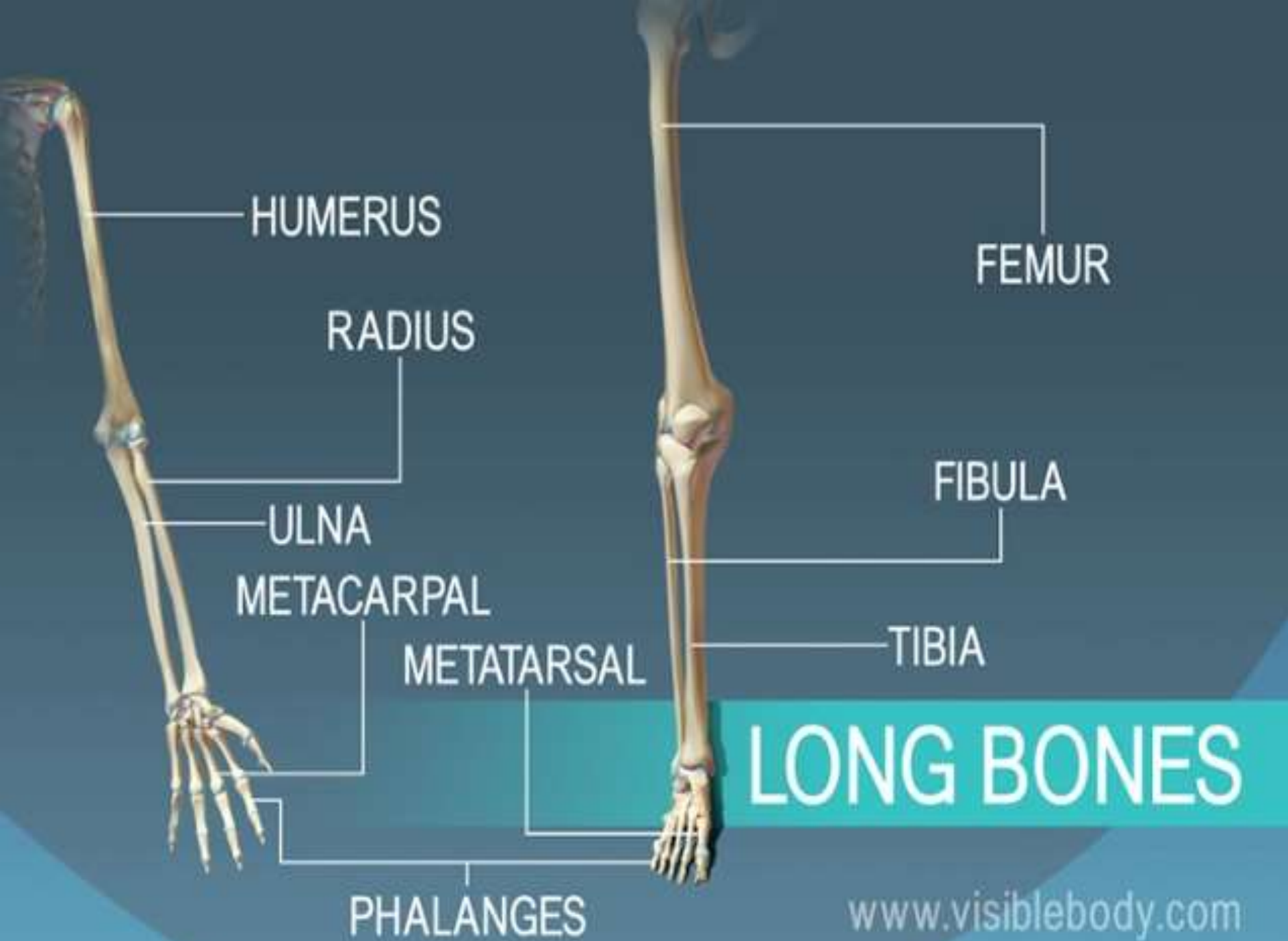
FLAT

SESAMOID

IRREGULAR

BONE TYPES

www.visiblebody.com



HUMERUS

RADIUS

ULNA

METACARPAL

METATARSAL

PHALANGES

FEMUR

FIBULA

TIBIA

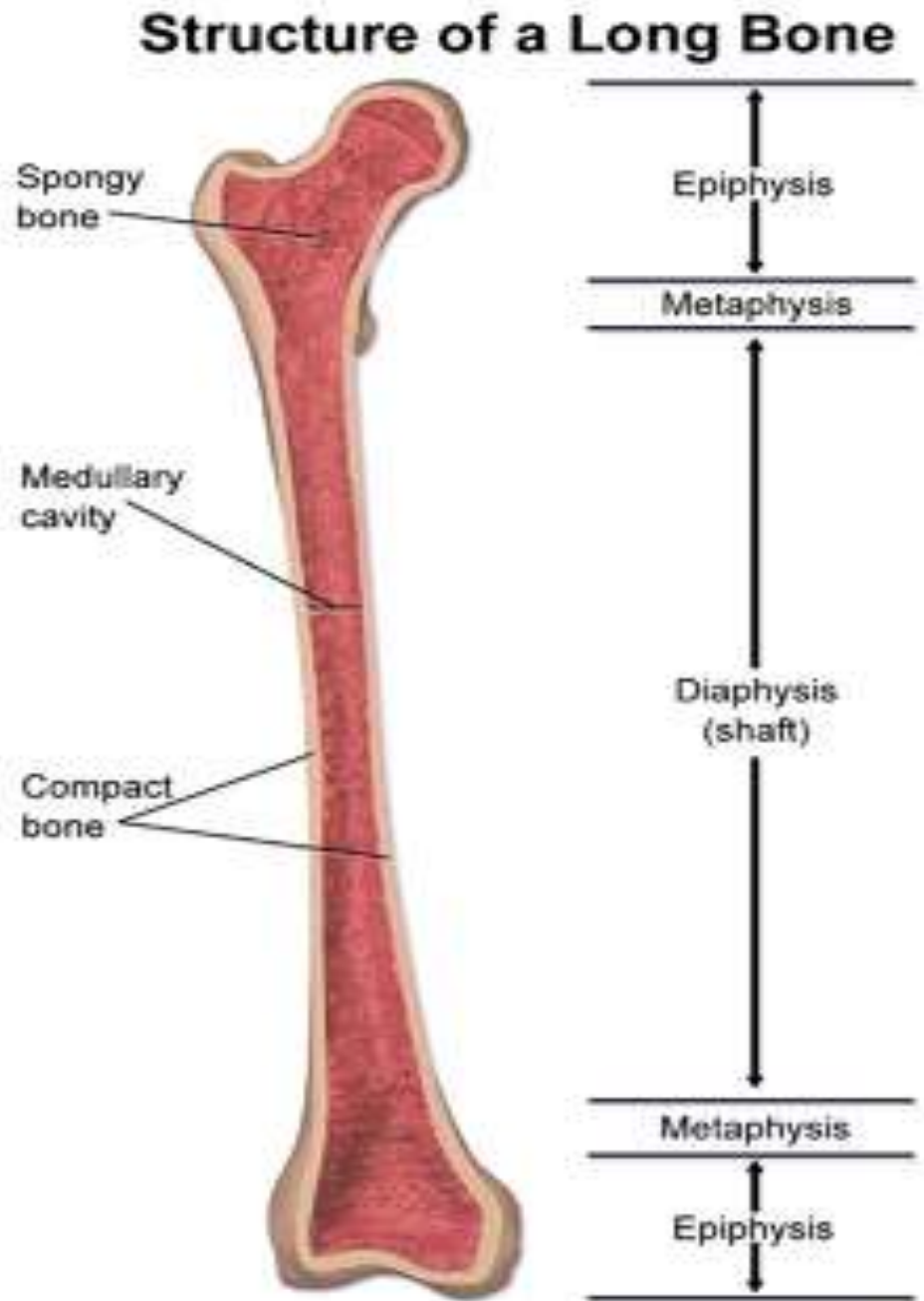
LONG BONES

Structure of long bone:

Epiphysis: ends

Diaphysis: shaft:
cortex and
medullary cavity

Periosteum: thick
fibrous tissue cover
the shaft

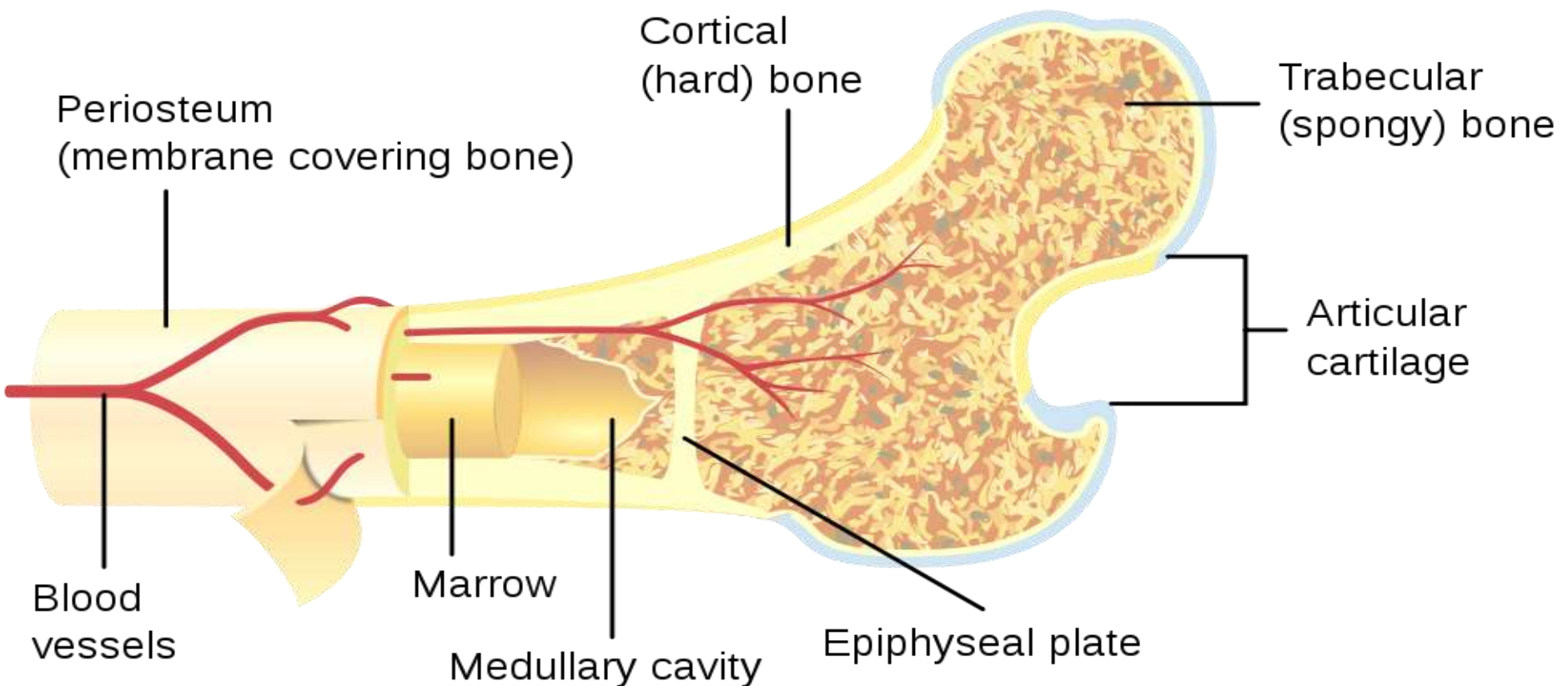


Structure of long bone:

Epiphysis: ends

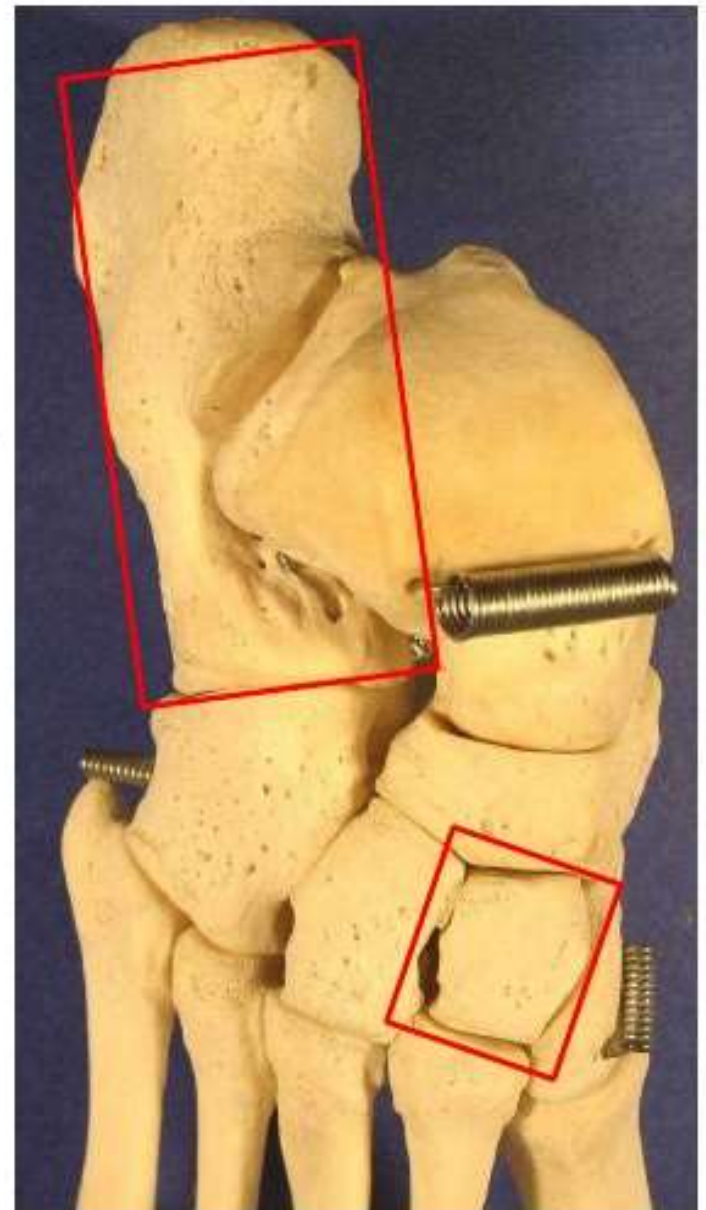
Diaphysis: shaft: cortex and medullary cavity

Periosteum: thick fibrous tissue cover the shaft

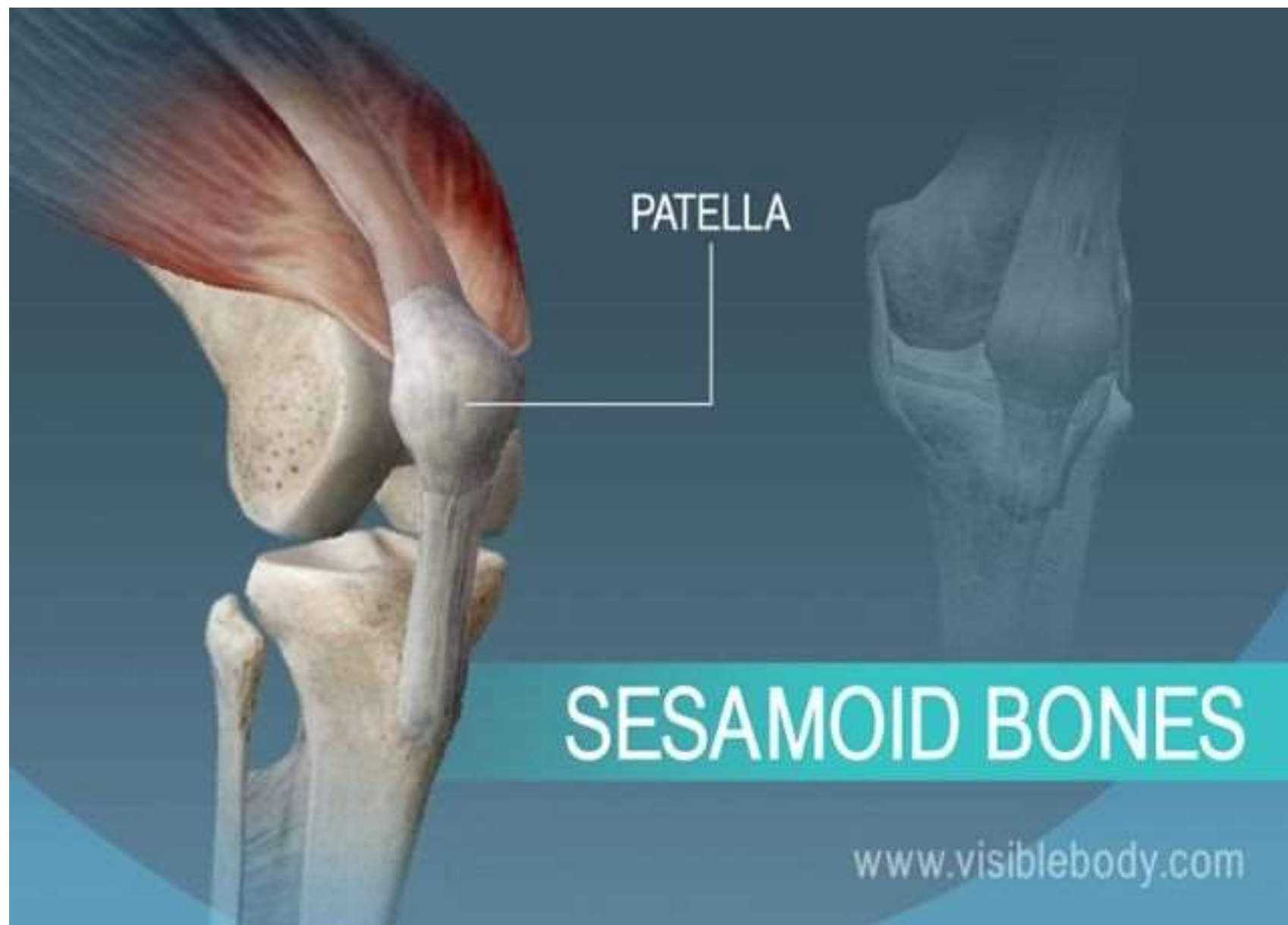


Short Bones

Short bones are roughly cube-shaped, or “cuboid,” meaning length, width, and height are all closer to equal, and they are associated with smaller, more complex movements.



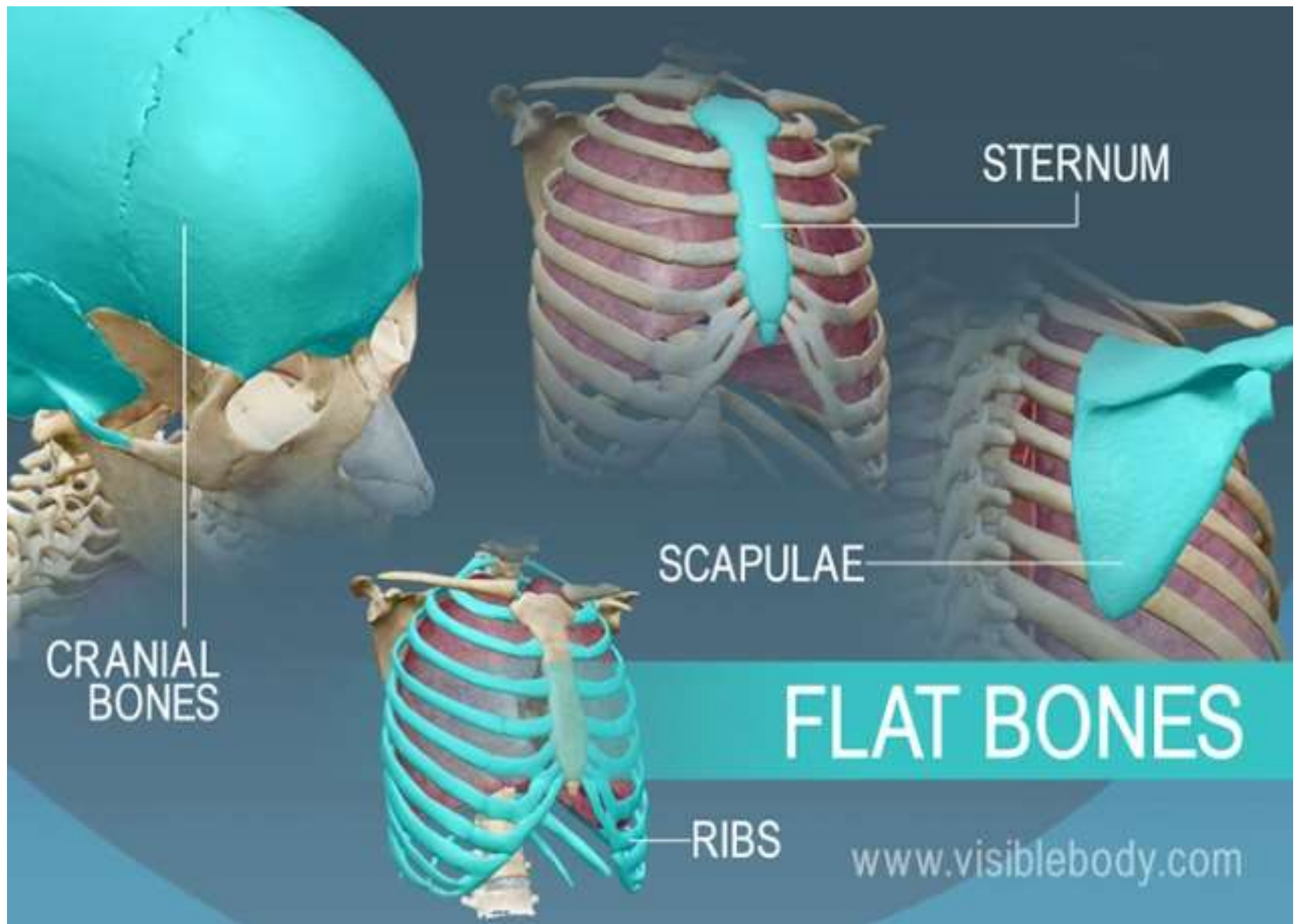
Note the “cuboid” shape



PATELLA

SESAMOID BONES

www.visiblebody.com



STERNUM

SCAPULAE

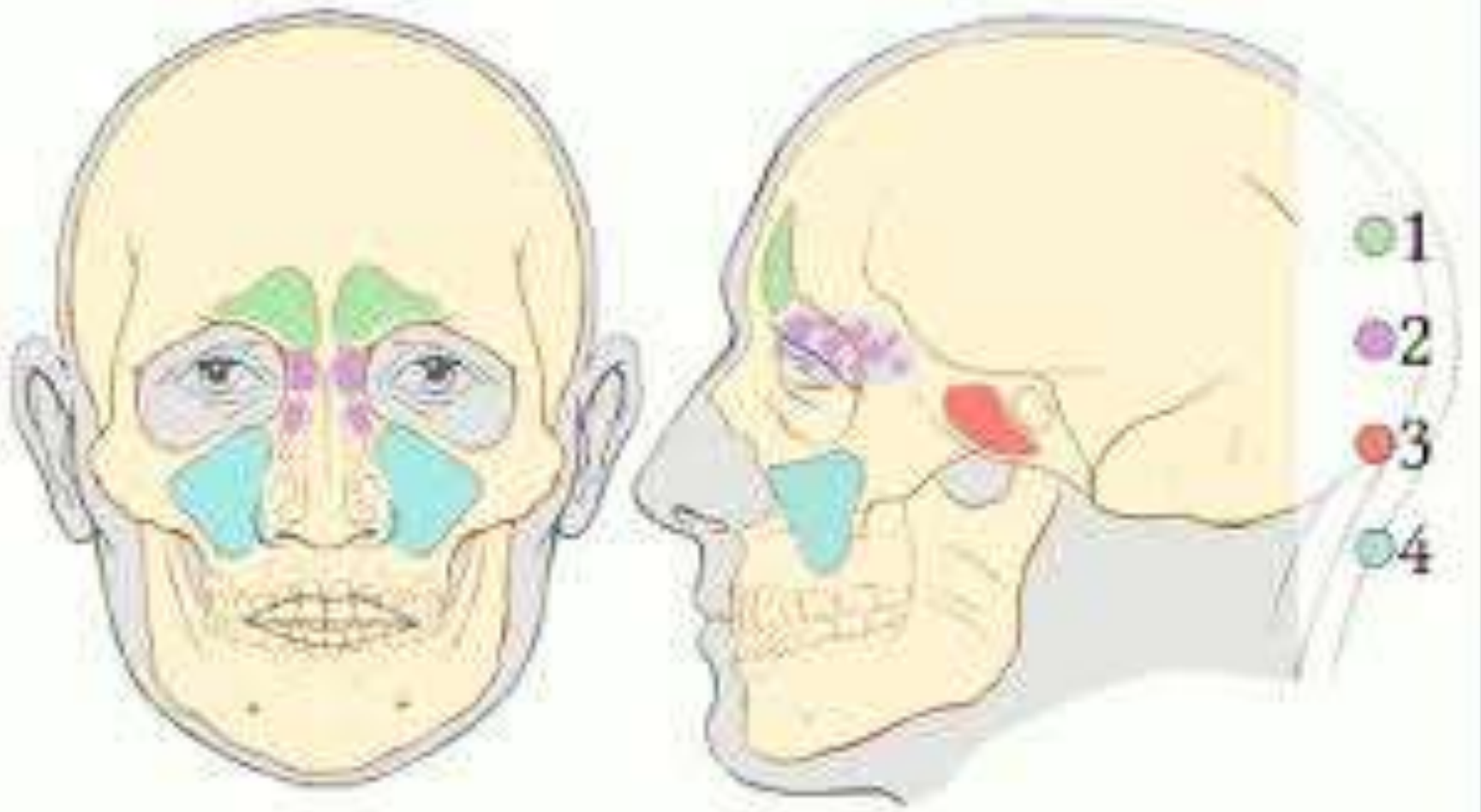
CRANIAL
BONES

RIBS

FLAT BONES

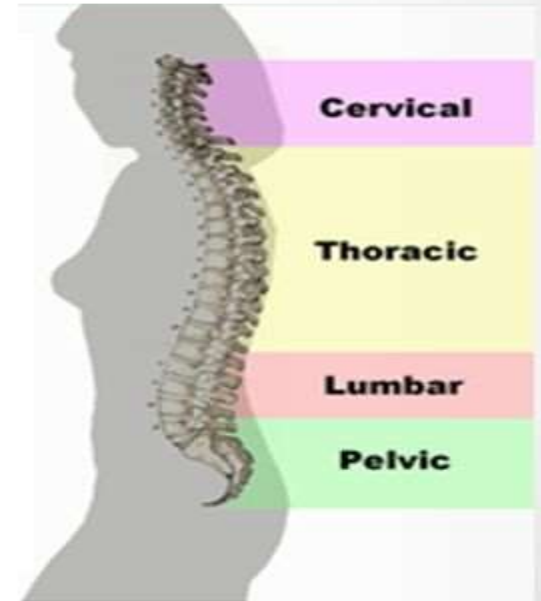
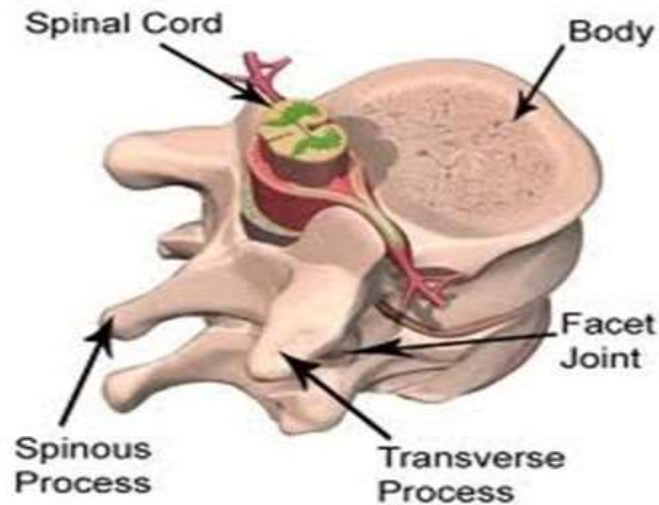
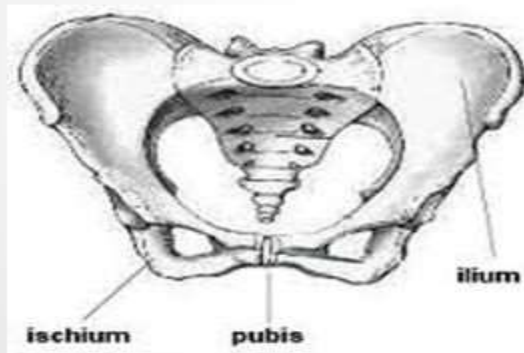
www.visiblebody.com

Pneumatic bones: The central part of the bone is absorbed. It is replaced by air-filled spaces
In skull : air sinuses



Irregular bones: Vertebrae and hip bones

Irregular Bones

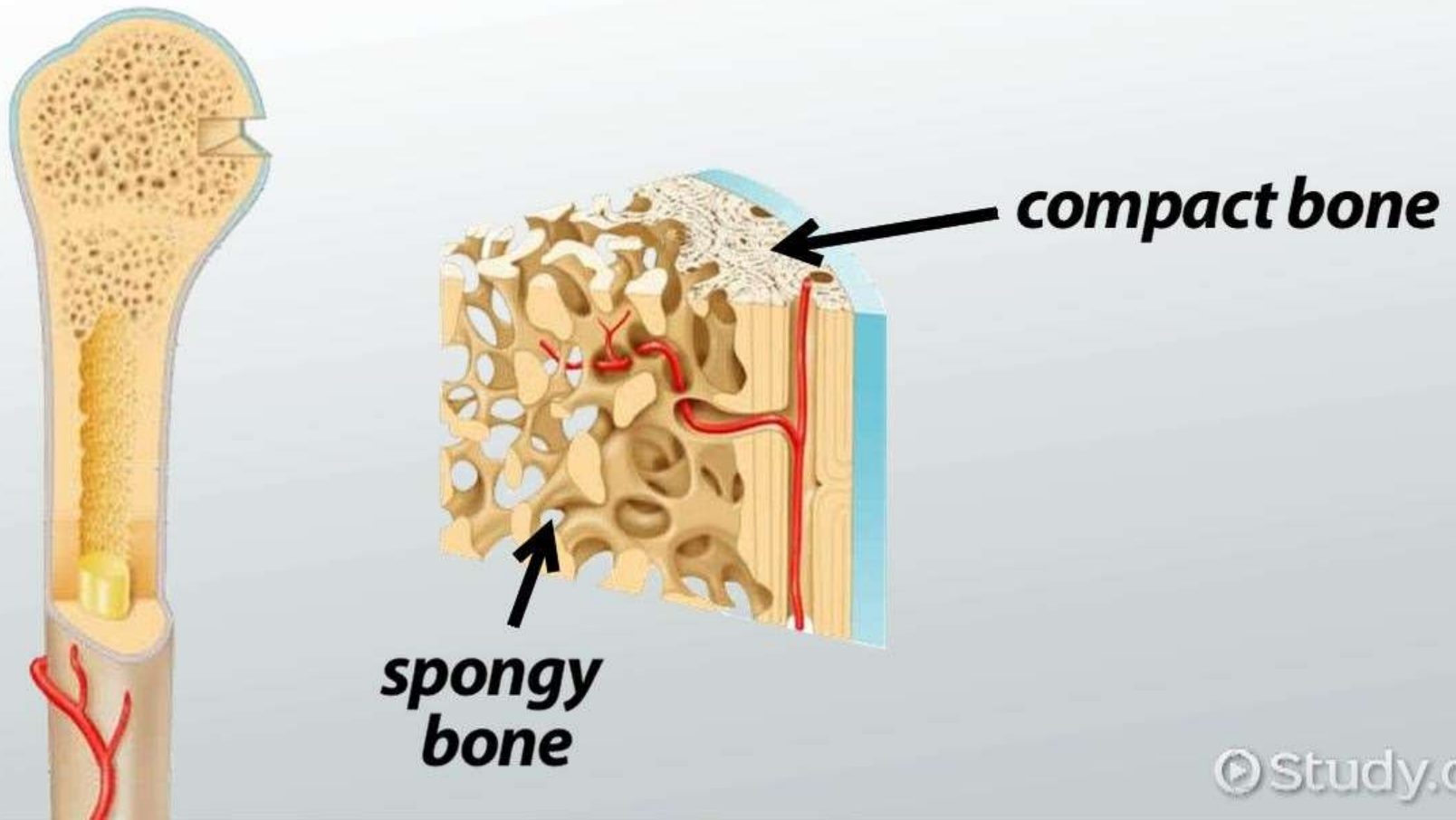


Types of bones according to the structure:

Compact bone: Outer layer. It is hard substance

Spongy bone: branching network of trabeculae with intervening spaces that contains red marrow

WHAT IS COMPACT BONE?



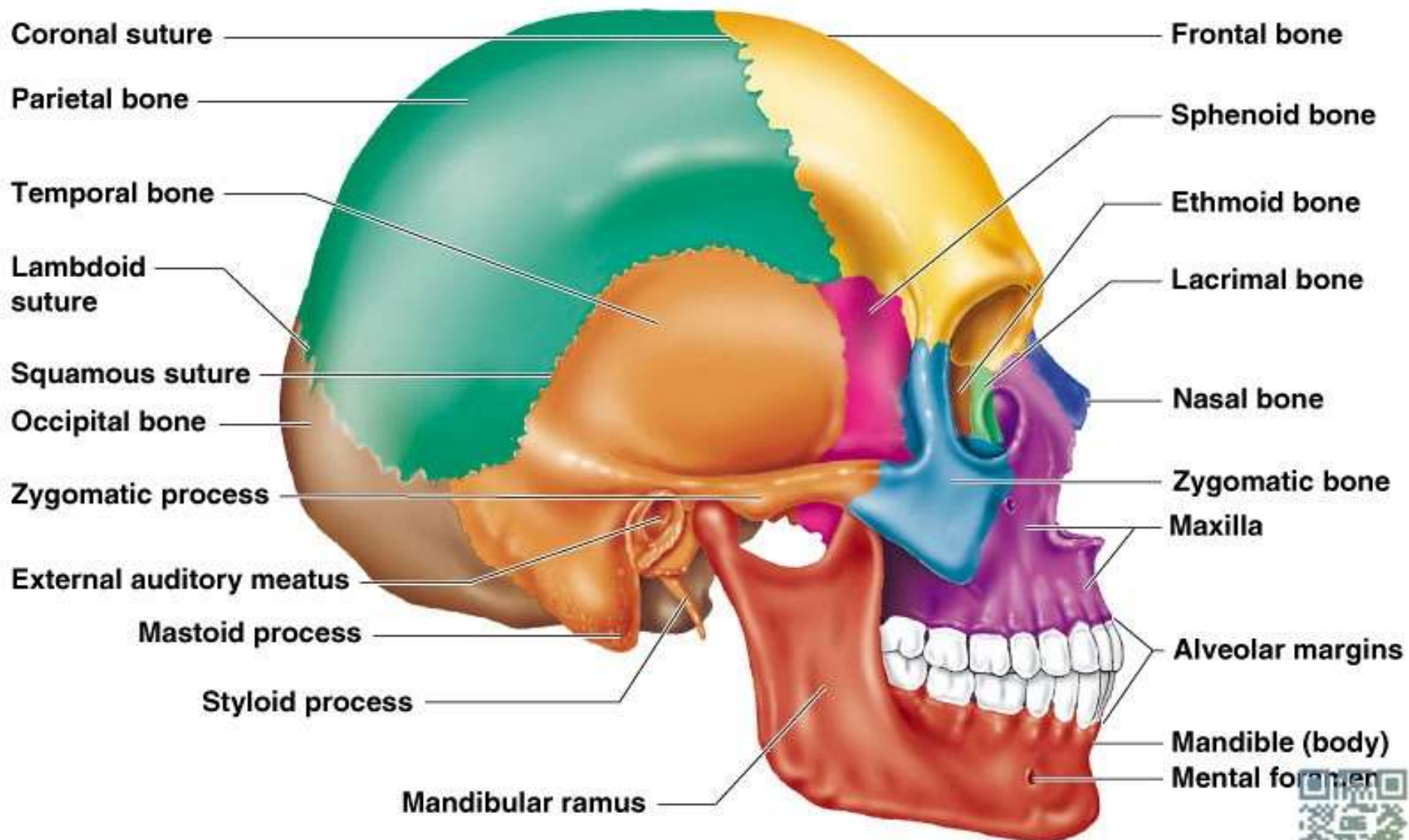
Bones according to region:

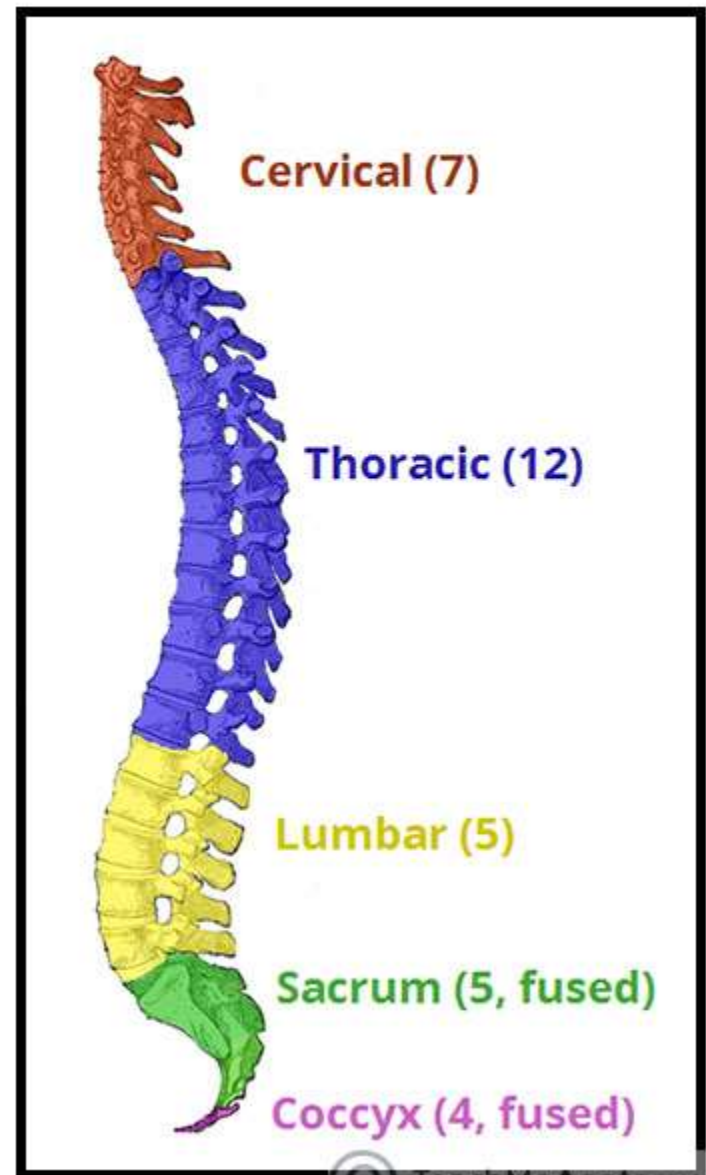
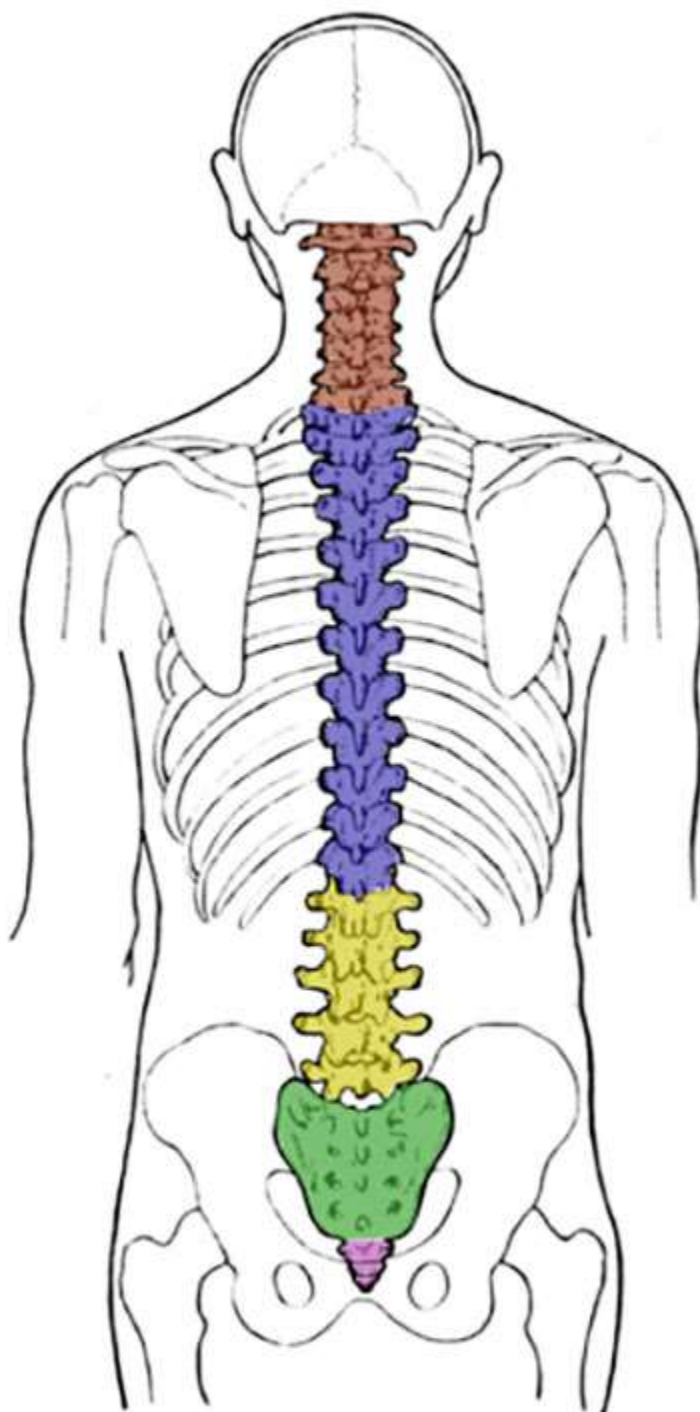
Appendicular Skeleton

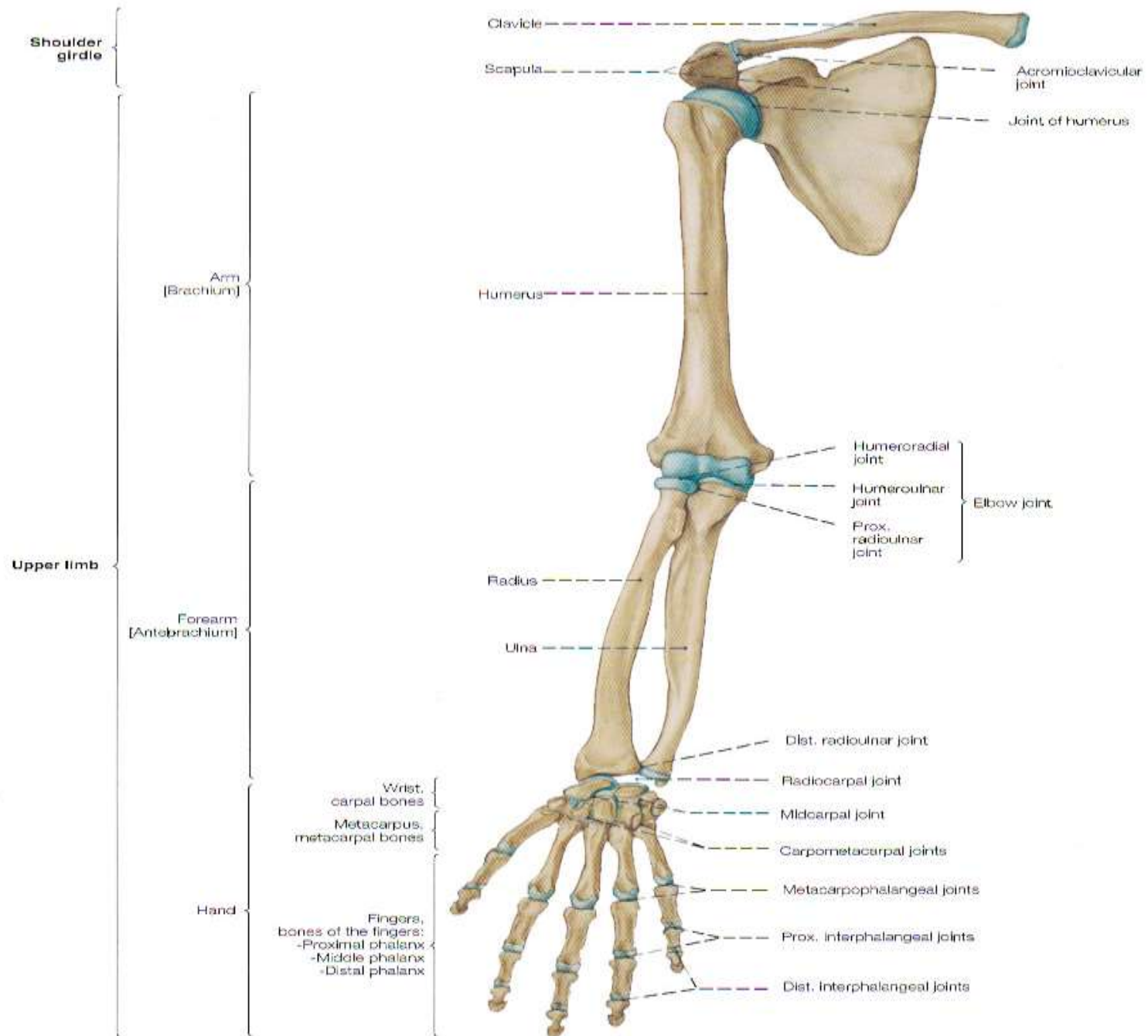


Axial Skeleton









Os Coxa

Femur

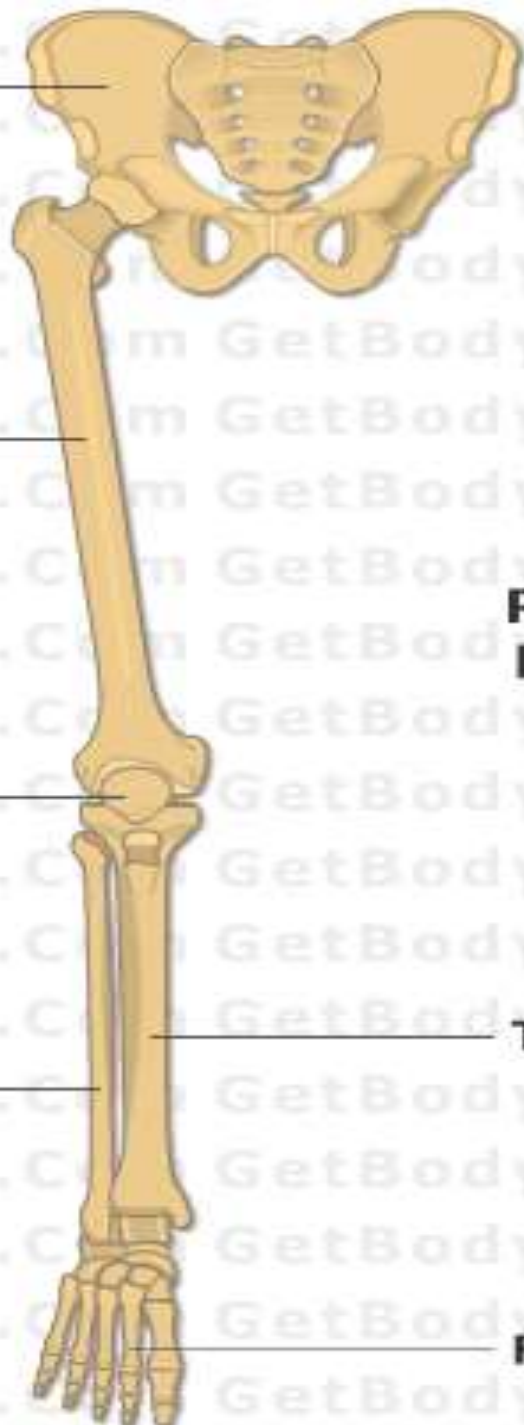
Patella

Fibula

**Right Lower
Limb Bones**

Tibia

Foot



Example of ball and socket joint is.....

Suture in skull is an example ofjoint

Femur isbone

.....is an example of flat bones

.....is an example of hinge joint

**..... is an example of secondary
cartilagenous joint**

.....is an irregular bone

*Thank
you*

